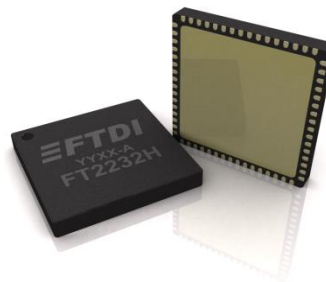


Future Technology Devices International Ltd

FT2232H

Dual High Speed USB to Multipurpose UART/FIFO IC

FT2232HQ-REEL, HongZhanpeng, 20241102



The FT2232H is FTDI's 5th generation of USB devices. The FT2232H is a USB 2.0 High Speed (480 Mb/s) to UART/FIFO IC. It has the capability of being configured in a variety of industry standard serial or parallel interfaces. The FT2232H has the following advanced features:

- Single chip USB to dual serial / parallel ports with a variety of configurations.
- Entire USB protocol handled on the chip. No USB specific firmware programming required.
- USB 2.0 High Speed (480 Mbits/Second) and Full Speed (12 Mbits/Second) compatible.
- Dual Multi-Protocol Synchronous Serial Engine (MPSSE) to simplify synchronous serial protocol (USB to JTAG, I²C, SPI or bit-bang) design.
- Dual independent UART or FIFO or MPSSE ports.
- Independent Baud rate generators.
- RS232/RS422/RS485 UART Transfer Data Rate up to 12 Mbaud. (RS232 Data Rate limited by external level shifter).
- USB to parallel FIFO transfer data rate up to 8 Mbyte/Sec.
- Single channel synchronous FIFO mode for transfers upto 40 Mbytes/Sec.
- CPU-style FIFO interface mode simplifies CPU interface design.
- MCU host bus emulation mode configuration option.
- Fast Opto-Isolated serial interface option.
- FTDI's royalty-free Virtual Com Port (VCP) and Direct (D2XX) drivers eliminate the requirement for USB driver development in most cases.
- Adjustable receive buffer timeout.
- Option for transmit and receive LED drive signals on each channel.
- Enhanced bit-bang Mode interface option with RD# and WR# strobes.
- FT245B-style FIFO interface option with bi-directional data bus and simple 4 wire handshake interface.
- Highly integrated design includes +1.8 V LDO regulator for V_{CORE}, integrated POR function and on chip clock multiplier PLL (12 MHz – 480 MHz).
- Asynchronous serial UART interface option with full hardware handshaking and modem interface signals.
- Fully assisted hardware or X-On / X-Off software handshaking.
- UART Interface supports 7/8 bit data, 1/2 stop bits, and Odd/Even/Mark/Space/No Parity.
- Auto-transmit enable control for RS485 serial applications using TXDEN pin.
- Operational configuration mode and USB Description strings configurable in external EEPROM over the USB interface.
- Configurable I/O drive strength (4, 8, 12 or 16 mA) and slew rate.
- Low operating and USB suspend current.
- Supports bus powered, self-powered and high-power bus powered USB configurations.
- UHCI/OHCI/EHCI host controller compatible.
- USB Bulk data transfer mode (512 byte packets in High Speed mode).
- +1.8 V (chip core) and +3.3 V I/O interfacing (+5 V Tolerant).
- Extended -40°C to 85°C industrial operating temperature range.
- Compact 64-LD Lead Free LQFP or QFN package.
- Available in compact Pb-free 56 Pin VQFN packages (RoHS compliant).
- +3.3 V single supply operating voltage range.
- ESD protection for FT2232H IO's:
Human Body Model (HBM) ±2 kV,
Machine Mode (MM) ±200 V,
Charge Device Model (CDM) ±500 V,
Latch-up free.

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1 Typical Applications

- Single chip USB to dual channel UART (RS232, RS422 or RS485).
- Single chip USB to dual channel FIFO.
- Single chip USB to dual channel JTAG.
- Single chip USB to dual channel SPI.
- Single chip USB to dual channel I2C.
- Single chip USB to dual channel Bit-Bang.
- Single chip USB to dual combination of any of above interfaces.
- Single chip USB to Fast Serial Optic Interface.
- Single chip USB to CPU target interface (as memory), double and independent.
- Single chip USB to Host Bus Emulation (as CPU).
- PDA to USB data transfer.
- USB Smart Card Readers.
- USB Instrumentation.
- USB Industrial Control
- USB MP3 Player Interface.
- USB FLASH Card Reader / Writers.
- Set Top Box PC - USB interface.
- USB Digital Camera Interface.
- USB Bar Code Readers.

1.1 Driver Support

Royalty free VIRTUAL COM PORT (VCP) and D2XX Direct Drivers are available for the following Operating Systems (OS):

- Windows
- Linux
- Mac
- Android (J2xx / D2xx only)

See the following website link for the full driver support list including OS versions and legacy OS.

<https://ftdichip.com/drivers/>

Virtual COM Port (VCP) drivers cause the USB device to appear as an additional COM port available to the PC. Application software can access the USB device in the same way as it would access a standard COM port.

D2XX Direct Drivers allow direct access to the USB device through a DLL. Application software can access the USB device through a series of DLL function calls. The functions available are listed in the [D2XX Programmer's Guide](#) document which is available from the [Documents](#) section of our website.

Please also refer to the Installation Guides webpage for details on how to install the drivers.

<https://ftdichip.com/document/installation-guides/>

The following additional installation guides, application notes and technical notes are also available:

- [AN_108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#)
- [AN_110 Programmers Guide for High-Speed FTCJTAG DLL](#)
- [AN_113 Interfacing FT2232H Hi-Speed Devices to I2C Bus](#)
- [AN_114 Interfacing FT2232H Hi-Speed Devices to SPI Bus](#)
- [AN_135 MPSSE Basics](#)
- [AN_177 User Guide for LibMPSSE-I2C](#)
- [AN_178 User Guide for LibMPSSE-SPI](#)
- [TN_104 Guide to Debugging Customers Failed Driver Installation](#)

1.2 Part Numbers

Part Number	Package
FT2232HL-xxxx	64 Pin LQFP
FT2232HQ-xxxx	64 Pin QFN
FT2232H-56Q-xxxx	56 Pin VQFN

Note: Packaging code for xxxx is:

- Reel: Taped and Reel (LQFP =1000 pcs per reel, QFN-64 =4000 pcs per reel, QFN-56 = 3000 pcs per reel)

- Tray: Tray packing, (LQFP =160 pcs per tray, QFN-64 =260 pcs per tray, QFN-56 = 260 pcs per tray)

Please refer to section 8 for all package mechanical parameters.

1.3 USB Compliant

The FT2232H is fully compliant with the USB 2.0 specification and has been given the USB-IF Test-ID (TID) 40720019.

The timing of the rise/fall time of the USB signals is not only dependant on the USB signal drivers, it is also dependant system and is affected by factors such as PCB layout, external components and any transient protection present on the USB signals. For USB compliance these may require a slight adjustment. This timing can be modified through a programmable setting stored in the same external EEPROM that is used for the USB descriptors. Timing can also be changed by adding appropriate passive components to the USB signals.



2 FT2232H Block Diagram

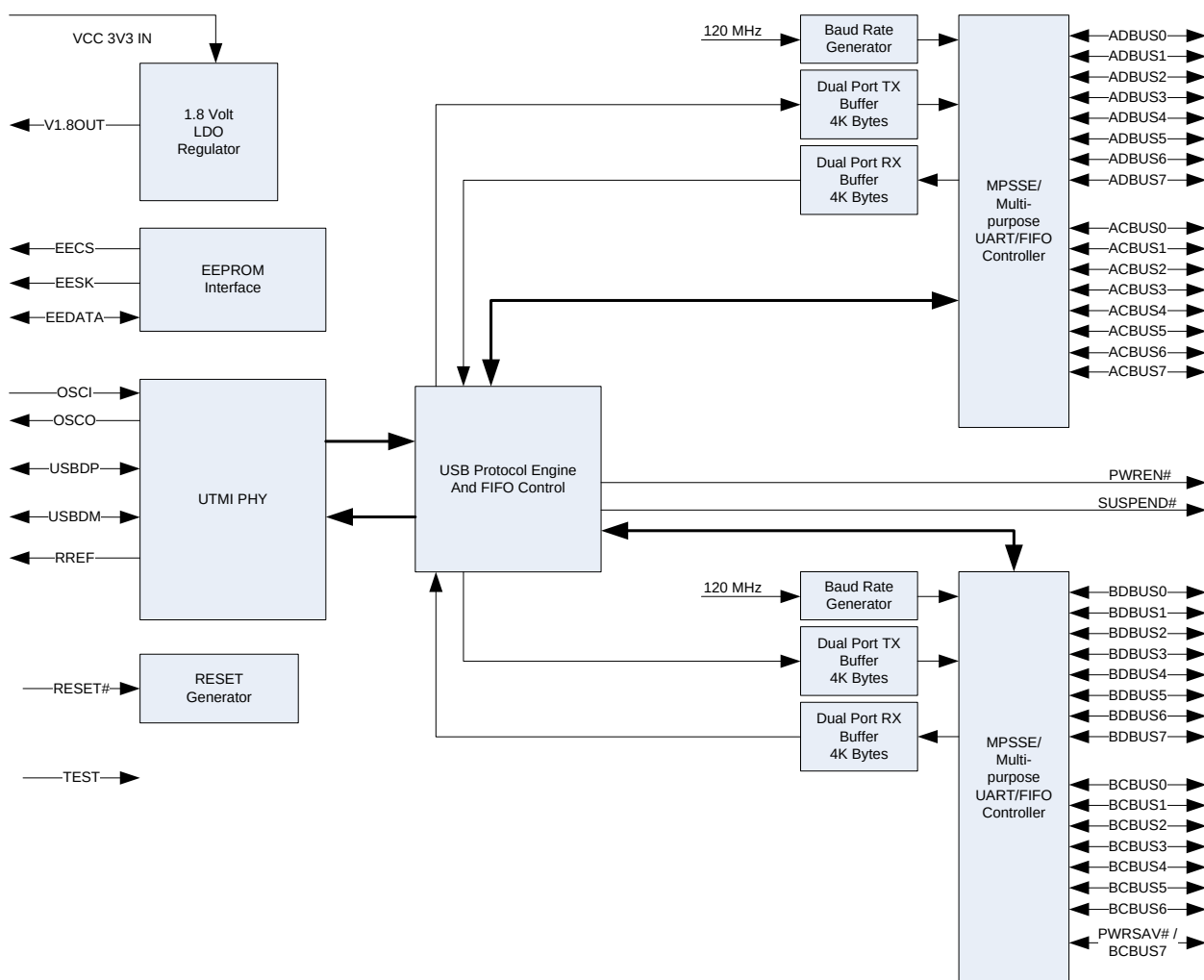


Figure 2.1 FT2232H Block Diagram

For a description of each function, please refer to Section 4.

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3 Device Pin Out and Signal Description

3.1 64-Pin LQFP and QFN Package

The 64-pin LQFP and 64-pin QFN have the same pin numbering for specific functions. This pin numbering is illustrated in the schematic symbol shown in Figure 3.1.

3.1.1 Schematic Symbol

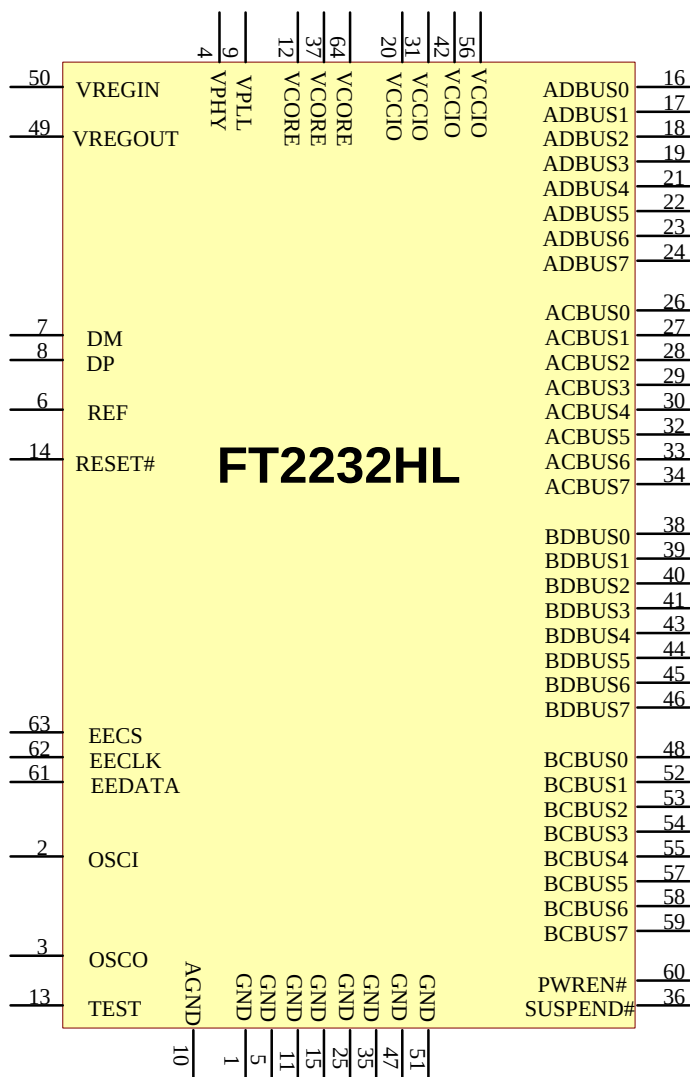


Figure 3.1 FT2232H Schematic Symbol

3.1.2 Pin Descriptions

This section describes the operation of the FT2232H pins for 64-pin LQFP and 64-pin QFN. Both the LQFP and the QFN packages have the same function on each pin. The function of many pins is determined by the configuration of the FT2232H. The following table details the function of each pin dependent on the configuration of the interface. Each of the functions is described in the following table (Note: The convention used throughout this document for active low signals is the signal name followed by #). Pins marked ** default to tri-stated inputs with an internal 75 K Ω (approx.) pull up resistor to VCCIO.

FT2232HL and FT2232HQ (64-pin)										
Pin		Pin functions (depends on configuration)								
Pin #	Pin Name	ASYNCR Serial (RS232)	245 FIFO SYNC	245 FIFO	ASYNCR Bit-bang	SYNCR Bit-bang	MPSSE	Fast Serial interface	CPU Style FIFO	Host Bus Emulation
Channel A										
16	ADBUS0	TXD	D0	D0	D0	D0	TCK/SK	USES	D0	AD0
17	ADBUS1	RXD	D1	D1	D1	D1	TDI/DO		D1	AD1
18	ADBUS2	RTS#	D2	D2	D2	D2	TDO/DI		D2	AD2
19	ADBUS3	CTS#	D3	D3	D3	D3	TMS/CS		D3	AD3
21	ADBUS4	DTR#	D4	D4	D4	D4	GPIOL0		D4	AD4
22	ADBUS5	DSR#	D5	D5	D5	D5	GPIOL1		D5	AD5
23	ADBUS6	DCD#	D6	D6	D6	D6	GPIOL2		D6	AD6
24	ADBUS7	RI#	D7	D7	D7	D7	GPIOL3		D7	AD7
26	ACBUS0	TXDEN	RXF#	RXF#	**	**	GPIOH0		CS#	A8
27	ACBUS1	**	TXE#	TXE#	WRSTB	WRSTB	GPIOH1		A0	A9
28	ACBUS2	**	RD#	RD#	RDSTB#	RDSTB#	GPIOH2		RD#	A10
29	ACBUS3	RXLED#	WR#	WR#	**	**	GPIOH3		WR#	A11
30	ACBUS4	TXLED#	SIWUA	SIWUA	SIWUA	SIWUA	GPIOH4		Note 1	A12
32	ACBUS5	**	CLKOUT	**	**	**	GPIOH5		**	A13
33	ACBUS6	**	OE#	**	**	**	GPIOH6		**	A14
34	ACBUS7	**	**	**	**	**	GPIOH7		**	A15
Channel B										
38	BDBUS0	TXD		D0	D0	D0	TCK/SK	FSDI	D0	CS#
39	BDBUS1	RXD		D1	D1	D1	TDI/DO	FSCLK	D1	ALE
40	BDBUS2	RTS#		D2	D2	D2	TDO/DI	FSDO	D2	RD#
41	BDBUS3	CTS#		D3	D3	D3	TMS/CS	FSCTS	D3	WR#
43	BDBUS4	DTR#		D4	D4	D4	GPIOL0		D4	IORDY
44	BDBUS5	DSR#		D5	D5	D5	GPIOL1		D5	CLKOUT
45	BDBUS6	DCD#		D6	D6	D6	GPIOL2		D6	I/O0
46	BDBUS7	RI#		D7	D7	D7	GPIOL3		D7	I/O1
48	BCBUS0	TXDEN		RXF#	**	**	GPIOH0		CS#	**
52	BCBUS1	**		TXE#	WRSTB	WRSTB	GPIOH1		A0	**
53	BCBUS2	**		RD#	RDSTB#	RDSTB#	GPIOH2		RD#	**
54	BCBUS3	RXLED#		WR#	**	**	GPIOH3		WR#	**
55	BCBUS4	TXLED#		SIWUB	SIWUB	SIWUB	GPIOH4	SIWUB	Note 1	**
57	BCBUS5	**		**	**	**	GPIOH5		**	**
58	BCBUS6	**		**	**	**	GPIOH6		**	**
59	BCBUS7	PWRSV	PWRSV	PWRSV	PWRSV	PWRSV	GPIOH7	PWRSV	PWRS	PWRSV
60	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWRE	PWREN#
36	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPEN D#	SUSPE ND#	SUSPEND #
Configuration memory interface										
63	EECS									
62	EECLK									
61	EEDATA									

Table 3.1 FT2232H Pin Configurations for 64-pin QFN and LQFP package

Note1: To wake up the USB device in this mode, put ACBUS0 from "High" to "LOW", set ACBUS1 "High", and put ACBUS3 from "High" to "LOW". The same also applies on the equivalent Channel B pins

Note: Initial Pin States - The device will start up as two UART ports. Therefore, pins which are output in UART mode will be driving out. If an application uses MPSSE or bit-bang, ensure that any external signals do not drive into these pins and cause contention until the application has configured the mode and direction of these lines. If a different mode other than UART is set in the EEPROM, the pins will change to the selected mode after the device has read the EEPROM.

3.1.3 Common Pins

The operation of the following FT2232H pins are the same regardless of the configured mode:-

Pin No.	Name	Type	Description
12,37,64	VCORE	POWER Input	+1.8 V input. Core supply voltage input.
20,31,42,56	VCCIO	POWER Input	+3.3 V input. I/O interface power supply input. Failure to connect all VCCIO pins will result in failure of the device.
9	VPLL	POWER Input	+3.3 V input. Internal PHY PLL power supply input. It is recommended that this supply is filtered using an LC filter.
4	VPHY	POWER Input	+3.3 V Input. Internal USB PHY power supply input. Note that this cannot be connected directly to the USB supply. A +3.3V regulator must be used. It is recommended that this supply is filtered using an LC filter.
50	VREGIN	POWER Input	+3.3 V Input. Integrated 1.8 V voltage regulator input.
49	VREGOUT	POWER Output	+1.8 V Output. Integrated voltage regulator output. Connect to VCORE with 3.3 μ F filter capacitor.
10	AGND	POWER Input	0 V Analog ground.
1,5,11,15,25,35,47,51	GND	POWER Input	0 V Ground input.

Table 3.2 Power and Ground for 64-pin QFN and LQFP package

Pin No.	Name	Type	Description
2	OSCI	INPUT	Oscillator input.
3	OSCO	OUTPUT	Oscillator output.
6	REF	INPUT	Current reference – connect via a 12 K Ω resistor @ 1% to GND.
7	DM	I/O	USB Data Signal Minus.
8	DP	I/O	USB Data Signal Plus.
13	TEST	INPUT	IC test pin – for normal operation should be connected to GND.
14	RESET#	INPUT	Reset input (active low).
60	PWREN#	OUTPUT	Active low power-enable output. PWREN# = 0: Normal operation. PWREN# = 1: USB SUSPEND mode or device has not been configured. This can be used by external circuitry to power down logic when device is in USB suspend or has not been configured.
36	SUSPEND#	OUTPUT	Active low when USB is in suspend mode.
59	PWRSV#	INPUT	USB Power Save input. This is an EEPROM configurable option used when the FT2232H is used in a self-powered mode and is used to prevent forcing current down the USB lines when the host or hub is powered off. PWRSV# = 1 : Normal Operation PWRSV# = 0: FT2232H forced into SUSPEND mode. PWRSV# can be connected to GND (via a 10 K Ω resistor) and another resistor (e.g. 4K7) connected to the VBUS of the USB connector. When this input goes high, then it indicates to the FT2232H that it is connected to a host PC. When the host or hub is powered down then the FT2232H is held in SUSPEND mode.

Table 3.3 Common Function pins for 64-pin QFN and LQFP package

Pin No.	Name	Type	Description
63	EECS	I/O	EEPROM – Chip Select. Tri-State during device reset.
62	EECLK	OUTPUT	Clock signal to EEPROM. Tri-State during device reset. When not in reset, this outputs the EEPROM clock.
61	EEDATA	I/O	EEPROM – Data I/O Connect directly to Data-In of the EEPROM and to Data-Out of the EEPROM via a 2.2 K resistor. Also, pull Data-Out of the EEPROM to VCC via a 10 K resistor for correct operation. Tri-State during device reset.

Table 3.4 EEPROM Interface Group for 64-pin QFN and LQFP package

3.1.4 Configured Pins

The following sections describe the function of the configurable pins referred to in Table 3.1 which is determined by how the FT2232H is configured.

3.1.4.1 FT2232H pins used in an RS232 Interface

The FT2232H channel A or channel B can be configured as an RS232 interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.5.

Channel A Pin No.	Channel B Pin No.	Name	Type	RS232 Configuration Description
16	38	TXD	OUTPUT	TXD = transmitter output
17	39	RXD	INPUT	RXD = receiver input
18	40	RTS#	OUTPUT	RTS# = Ready To send handshake output
19	41	CTS#	INPUT	CTS# = Clear To Send handshake input
21	43	DTR#	OUTPUT	DTR# = Data Transmit Ready modem signaling line
22	44	DSR#	INPUT	DSR# = Data Set Ready modem signaling line
23	45	DCD#	INPUT	DCD# = Data Carrier Detect modem signaling line
24	46	RI#	INPUT	RI# = Ring Indicator Control Input. When the Remote Wake up option is enabled in the EEPROM, taking RI# low can be used to resume the PC USB Host controller from suspend. (Also see note 1, 2, 3 in section 4.12)
26	48	TXDEN	OUTPUT	TXDEN = (TTL level). For use with RS485 level converters.
29	54	RXLED#	OUTPUT	RXLED = Receive signaling output when data is transferred from FT2232H to USB Host. Pulses low when receiving data (RXD) via USB. This should be connected to an LED.
30	55	TXLED#	OUTPUT	TXLED = Transmit signaling output when data is transferred from USB Host to FT2232H. Pulses low when transmitting data (TXD) via USB. This should be connected to an LED.

Table 3.5 Channel A and Channel B RS232 Configured Pin Descriptions

3.1.4.2 FT2232H pins used in an FT245 Style Synchronous FIFO Interface

The FT2232H only channel A can be configured as a FT245 style synchronous FIFO interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.6. To enter this mode the external EEPROM must be set to make port A 245 mode. A software command (Set Bit Mode option) is then sent by the application to the FTDI driver to tell the chip to enter single channel synchronous FIFO mode. In this mode the 'B' channel is not available as all resources have been switched onto channel A. In this mode, data is written or read on the rising edge of the CLKOUT.

Channel A Pin No.	Name	Type	FT245 Configuration Description
24,23,22,21,19,18,17,16	ADBUS[7:0]	I/O	D7 to D0 bidirectional FIFO data. This bus is normally input unless OE# is low.
26	RXF#	OUTPUT	When high, do not read data from the FIFO. When low, there is data available in the FIFO which can be read by driving RD# low. When in synchronous mode, data is transferred on every clock that RXF# and RD# are both low. Note that the OE# pin must be driven low at least 1 clock period before asserting RD# low.
27	TXE#	OUTPUT	When high, do not write data into the FIFO. When low, data can be written into the FIFO by driving WR# low. When in synchronous mode, data is transferred on every clock that TXE# and WR# are both low.
28	RD#	INPUT	Enables the current FIFO data byte to be driven onto D0...D7 when RD# goes low. The next FIFO data byte (if available) is fetched from the receive FIFO buffer each CLKOUT cycle until RD# goes high.
29	WR#	INPUT	Enables the data byte on the D0...D7 pins to be written into the transmit FIFO buffer when WR# is low. The next FIFO data byte is written to the transmit FIFO buffer each CLKOUT cycle until WR# goes high.
32	CLKOUT	OUTPUT	60 MHz Clock driven from the chip. All signals should be synchronized to this clock.
33	OE#	INPUT	Output enable when low to drive data onto D0-7. This should be driven low at least 1 clock period before driving RD# low to allow for data buffer turn-around.
30	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# = 0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.6 Channel A FT245 Style Synchronous FIFO Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.4.

3.1.4.3 FT2232H pins used in an FT245 Style Asynchronous FIFO Interface

The FT2232H channel A or channel B can be configured as a FT245 asynchronous FIFO interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.7. To enter this mode the external EEPROM must be set to make port A or B or both 245 mode. In this mode, data is written or read on the falling edge of the RD# or WR# signals.

Channel A Pin No.	Channel B Pin No.	Name	Type	FT245 Configuration Description
24,23,22,21,19,18,17,16	46,45,44,43,41,40,39,38	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional FIFO data. This bus is normally input unless RD# is low.
26	48	RXF#	OUTPUT	When high, do not read data from the FIFO.

Channel A Pin No.	Channel B Pin No.	Name	Type	FT245 Configuration Description
				When low, there is data available in the FIFO which can be read by driving RD# low. When RD# goes high again RXF# will always go high and only become low again if there is another byte to read. During reset this signal pin is tri-state, but pulled up to VCCIO via an internal 200 KΩ resistor.
27	52	TXE#	OUTPUT	When high, do not write data into the FIFO. When low, data can be written into the FIFO by strobing WR# high, then low. During reset this signal pin is tri-state, but pulled up to VCCIO via an internal 200 KΩ resistor.
28	53	RD#	INPUT	Enables the current FIFO data byte to be driven onto D0...D7 when RD# goes low. Fetches the next FIFO data byte (if available) from the receive FIFO buffer when RD# goes high.
29	54	WR#	INPUT	Writes the data byte on the D0...D7 pins into the transmit FIFO buffer when WR# goes from high to low.
30	55	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# = 0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.7 Channel A and Channel B FT245 Style Asynchronous FIFO Configured Pin Descriptions

3.1.4.4 FT2232H pins used in a Synchronous or Asynchronous Bit-Bang Interface

The FT2232H channel A or channel B can be configured as a synchronous or asynchronous bit-bang interface. Bit-bang mode is a special FTDI FT2232H device mode that changes the 8 IO lines on either (or both) channels into an 8 bit bi-directional data bus. There are two types of bit-bang modes: synchronous and asynchronous.

When configured in any bit-bang mode, the pins used and the descriptions of the signals are shown in Table 3.8.

Channel A Pin No.	Channel B Pin No.	Name	Type	Configuration Description
24,23,22,21,19,18,17,16	46,45,44,43,41,40,39,38	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional Bit-Bang parallel I/O data pins
27	52	WRSTB#	OUTPUT	Write strobe, active low output indicates when new data has been written to the I/O pins from the Host PC (via the USB interface).
28	53	RDSTB#	OUTPUT	Read strobe, this output rising edge indicates when data has been read from the parallel I/O pins and sent to the Host PC (via

Channel A Pin No.	Channel B Pin No.	Name	Type	Configuration Description
				the USB interface).
30	55	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# = 0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.8 Channel A and Channel B Synchronous or Asynchronous Bit-Bang Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.10.

3.1.4.5 FT2232H pins used in an MPSSE

The FT2232H channel A and channel B, each have a Multi-Protocol Synchronous Serial Engine (MPSSE). Each MPSSE can be independently configured to a number of industry standard serial interface protocols such as JTAG, I2C or SPI, or it can be used to implement a proprietary bus protocol. For example, it is possible to use one of the FT2232H's channels to connect to an SRAM configurable FPGA such as supplied by Altera or Xilinx. The FPGA device would normally be un-configured (i.e. have no defined function) at power-up. Application software on the PC could use the MPSSE to download configuration data to the FPGA over USB. This data would define the hardware function on power up. The other FT2232H channel would be available for another function. Alternatively each MPSSE can be used to control a number of GPIO pins. When configured in this mode, the pins used and the descriptions of the signals are shown Table 3.9.

Channel A Pin No.	Channel B Pin No.	Name	Type	MPSSE Configuration Description
16	38	TCK/SK	OUTPUT	Clock Signal Output. For example: JTAG – TCK, Test interface clock SPI – SK, Serial Clock
17	39	TDI/DO	OUTPUT	Serial Data Output. For example: JTAG – TDI, Test Data Input SPI – DO
18	40	TDO/DI	INPUT	Serial Data Input. For example: JTAG – TDO, Test Data output SPI – DI, Serial Data Input
19	41	TMS/CS	OUTPUT	Output Signal Select. For example: JTAG – TMS, Test Mode Select SPI – CS, Serial Chip Select
21	43	GPIOL0	I/O	General Purpose input/output
22	44	GPIOL1	I/O	General Purpose input/output
23	45	GPIOL2	I/O	General Purpose input/output
24	46	GPIOL3	I/O	General Purpose input/output

Channel A Pin No.	Channel B Pin No.	Name	Type	MPSSE Configuration Description
26	48	GPIOH0	I/O	General Purpose input/output
27	52	GPIOH1	I/O	General Purpose input/output
28	53	GPIOH2	I/O	General Purpose input/output
29	54	GPIOH3	I/O	General Purpose input/output
30	55	GPIOH4	I/O	General Purpose input/output
32	57	GPIOH5	I/O	General Purpose input/output
33	58	GPIOH6	I/O	General Purpose input/output
34	59	GPIOH7	I/O	General Purpose input/output

Table 3.9 Channel A and Channel B MPSSE Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.6.

3.1.4.6 FT2232H Pins used as a Fast Serial Interface

The FT2232H channel B can be configured for use with high-speed optical bi-directional isolated serial data transfer: Fast Serial Interface. (Not available on channel A). A proprietary FTDI protocol designed to allow galvanic isolated devices to communicate synchronously with the FT2232H using just 4 signal wires (over two dual opto-isolators), and two power lines. The peripheral circuitry controls the data transfer rate in both directions, whilst maintaining full data integrity. Maximum USB full speed data rates can be achieved. Both 'A' and 'B' channels can communicate over the same 4 wire interface if desired.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.10.

Channel B Pin No.	Name	Type	Fast Serial Interface Configuration Description
38	FSDI	INPUT	Fast serial data input.
39	FSCLK	INPUT	Fast serial clock input. Clock input to FT2232H chip to clock data in or out.
40	FSDO	OUTPUT	Fast serial data output.
41	FSCTS	OUTPUT	Fast serial Clear To Send signal output. Driven low to indicate that the chip is ready to send data

Table 3.10 Channel B Fast Serial Interface Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.8.

3.1.4.7 FT2232H Pins Configured as a CPU-style FIFO Interface

The FT2232H channel A or channel B can be configured in a CPU-style FIFO interface mode which allows a CPU to interface to USB via the FT2232H. This mode is enabled in the external EEPROM.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.11.

Channel A Pin No.	Channel B Pin No.	Name	Type	Fast Serial Interface Configuration Description
24,23,22,21,19,18,17,16	46,45,44,43,41,40,39,38	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional data bus
26	48	CS#	INPUT	Active low chip select input
27	52	A0	INPUT	Address bit A0
28	53	RD#	INPUT	Active Low FIFO Read input
29	54	WR#	INPUT	Active Low FIFO Write input

Table 3.11 Channel A and Channel B CPU-style FIFO Interface Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.9.

3.1.4.8 FT2232H Pins Configured as a Host Bus Emulation Interface

The FT2232H can be used to combine channel A and channel B to be configured as a host bus emulation interface mode which emulates a standard 8048 or 8051 MCU host.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.12

Pin No.	Name	Type	Fast Serial Interface Configuration Description
24,23,22,21 19,18,17,16	ADBUS[7:0]	I/O	Multiplexed bidirectional Address/Data bus AD7 to AD0
34,33,32,30 29,28,27,26	A[15:8]	OUTPUT	Extended Address A15 to A8
38	CS#	OUTPUT	Active low chip select device during Read or Write.
39	ALE	OUTPUT	Positive pulse to latch the address
40	RD#	OUTPUT	Active low read output.
41	WR#	OUTPUT	Active low write output. (Data is setup before WR# goes low, and is held after WR# goes high)
43	IORDY	INPUT	Extends the time taken to perform a Read or Write operation if driven low. Pull up to V _{CORE} if not being used.
44	CLKOUT	OUTPUT	Master clock. Outputs the clock signal being used by the configured interface.
45	I/O0	I/O	MPSSE mode instructions to set / clear or read the high byte of data can be used with this pin. Please refer to application note AN_108 Command Processor For MPSSE and MCU Host Bus Emulation Modes for operation of these instructions.
46	I/O1	I/O	MPSSE mode instructions to set / clear or read the high byte of data can be used with this pin. In addition this pin has instructions which will make the controller wait until it is high, or wait until it is low. This can be used to connect to an IRQ pin of a peripheral chip. The FT2232H will wait for the interrupt, and then read the device, and pass the answer back to the host PC. I/O1 must be held in input mode if this option is used. Please refer to application note AN_108 Command Processor For MPSSE and MCU Host Bus Emulation Modes for operation of these instructions.

Table 3.12 Channel A and Channel B Host Bus Emulation Interface Configured Pin Descriptions

For a functional description of this mode, please refer to Section 4.7.

3.2 56-Pin VQFN Package

The 56-pin VQFN with lower pin count and small size package is also available for FT2232H. The differences exist on power/ground and pin number for each pin. The part number is as FT2232H-56Q to distinguish from the 64-pin package type. All the functions are supported in the 56-pin VQFN package. The pin numbering is illustrated in the schematic symbol shown in Figure 3.2.

3.2.1 Schematic Symbol for FT2232H-56Q

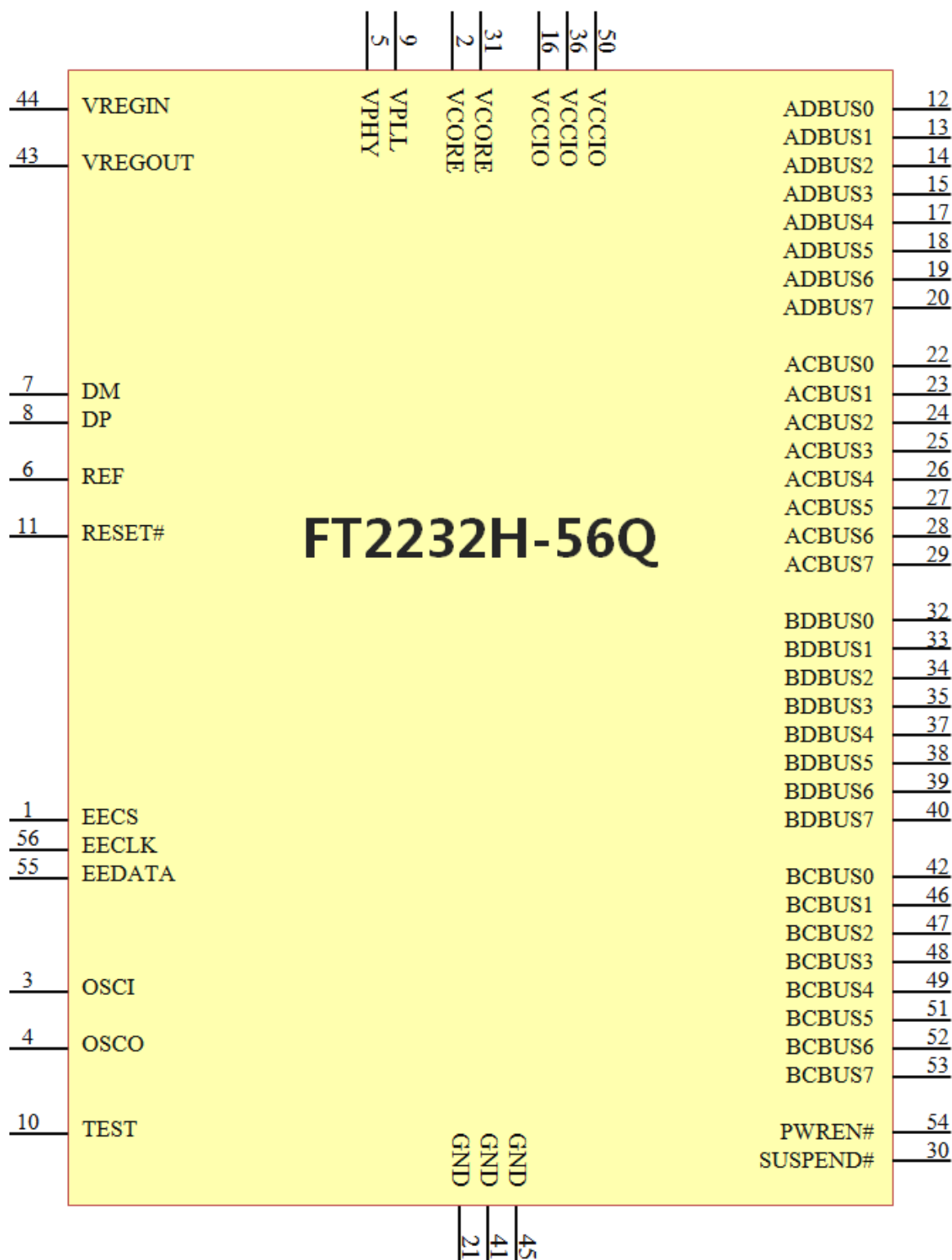


Figure 3.2 FT2232H-56Q Schematic Symbol

3.2.2 Pin Descriptions for FT2232H-56Q

This section describes the operation of the FT2232H-56Q pins for 56-pin VQFN package. The function of many pins is determined by the configuration of the FT2232H-56Q. The following table details the function of each pin dependent on the configuration of the interface. Each of the functions is described in

Table 3.13. (Note: The convention used throughout this document for active low signals is the signal name followed by #). Pins marked ** default to tri-stated inputs with an internal 75 K Ω (approx.) pull up resistor to VCCIO.

FT2232H-56Q										
Pin		Pin functions (depends on configuration)								
Pin #	Pin Name	ASYNCRS232	245FIFOSYNCRS232	245FIFOSYNCRS232	ASYNCRS232	SYNCRS232	MPSSE	Fast Serial interface	CPU Style FIFO	Host Bus Emulation
Channel A										
12	ADBUS0	TXD	D0	D0	D0	D0	TCK/SK	USES	D0	AD0
13	ADBUS1	RXD	D1	D1	D1	D1	TDI/DO		D1	AD1
14	ADBUS2	RTS#	D2	D2	D2	D2	TDO/DI		D2	AD2
15	ADBUS3	CTS#	D3	D3	D3	D3	TMS/CS		D3	AD3
17	ADBUS4	DTR#	D4	D4	D4	D4	GPIOL0		D4	AD4
18	ADBUS5	DSR#	D5	D5	D5	D5	GPIOL1		D5	AD5
19	ADBUS6	DCD#	D6	D6	D6	D6	GPIOL2		D6	AD6
20	ADBUS7	RI#	D7	D7	D7	D7	GPIOL3		D7	AD7
22	ACBUS0	TXDEN	RXF#	RXF#	**	**	GPIOH0		CS#	A8
23	ACBUS1	**	TXE#	TXE#	WRSTB	WRSTB	GPIOH1		A0	A9
24	ACBUS2	**	RD#	RD#	RDSTB#	RDSTB#	GPIOH2		RD#	A10
25	ACBUS3	RXLED#	WR#	WR#	**	**	GPIOH3		WR#	A11
26	ACBUS4	TXLED#	SIWUA	SIWUA	SIWUA	SIWUA	GPIOH4		Note 1	A12
27	ACBUS5	**	CLKOUT	**	**	**	GPIOH5		**	A13
28	ACBUS6	**	OE#	**	**	**	GPIOH6		**	A14
29	ACBUS7	**	**	**	**	**	GPIOH7		**	A15
Channel B										
32	BDBUS0	TXD		D0	D0	D0	TCK/SK	FSDI	D0	CS#
33	BDBUS1	RXD		D1	D1	D1	TDI/DO	FSCLK	D1	ALE
34	BDBUS2	RTS#		D2	D2	D2	TDO/DI	FSDO	D2	RD#
35	BDBUS3	CTS#		D3	D3	D3	TMS/CS	FSCTS	D3	WR#
37	BDBUS4	DTR#		D4	D4	D4	GPIOL0		D4	IORDY
38	BDBUS5	DSR#		D5	D5	D5	GPIOL1		D5	CLKOUT
39	BDBUS6	DCD#		D6	D6	D6	GPIOL2		D6	I/O0
40	BDBUS7	RI#		D7	D7	D7	GPIOL3		D7	I/O1
42	BCBUS0	TXDEN		RXF#	**	**	GPIOH0		CS#	**
46	BCBUS1	**		TXE#	WRSTB	WRSTB	GPIOH1		A0	**
47	BCBUS2	**		RD#	RDSTB#	RDSTB#	GPIOH2		RD#	**
48	BCBUS3	RXLED#		WR#	**	**	GPIOH3		WR#	**
49	BCBUS4	TXLED#		SIWUB	SIWUB	SIWUB	GPIOH4	SIWUB	Note 1	**
51	BCBUS5	**		**	**	**	GPIOH5		**	**
52	BCBUS6	**		**	**	**	GPIOH6		**	**
53	BCBUS7	PWRSV	PWRSV	PWRSV	PWRSV	PWRSV	GPIOH7	PWRSV	PWRSV	PWRSV
54	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN	PWREN
30	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#	SUSPEND#
Configuration memory interface										
1	EECS									
56	EECLK									
55	EEDATA									

Table 3.13 FT2232H Pin Configurations for 56-pin VQFN package

Note1: To wake up the USB device in this mode, put ACBUS0 from "High" to "LOW", set ACBUS1 "High", and put ACBUS3 from "High" to "LOW". The same also applies on the equivalent Channel B pins.

NOTE: Initial Pin States - The device will start up as two UART ports. Therefore pins which are output in UART mode will be driving out. If an application uses MPSSE or bit-bang, ensure that any external signals do not drive into these pins and cause contention until the application has configured the mode and direction of these lines. If a different mode other than UART is set in the EEPROM, the pins will change to the selected mode after the device has read the EEPROM.

3.2.3 Common Pins for FT2232H-56Q

The operation of the following FT2232H-56Q pins are the same regardless of the configured mode:-

Pin No.	Name	Type	Description
2,31	VCORE	POWER Input	+1.8 V input. Core supply voltage input.
16,36,50	VCCIO	POWER Input	+3.3 V input. I/O interface power supply input. Failure to connect all VCCIO pins will result in failure of the device.
9	VPLL	POWER Input	+3.3 V input. Internal PHY PLL power supply input. It is recommended that this supply is filtered using an LC filter.
5	VPHY	POWER Input	+3.3 V Input. Internal USB PHY power supply input. Note that this cannot be connected directly to the USB supply. A +3.3 V regulator must be used. It is recommended that this supply is filtered using an LC filter.
44	VREGIN	POWER Input	+3.3 V Input. Integrated 1.8 V voltage regulator input.
43	VREGOUT	POWER Output	+1.8 V Output. Integrated voltage regulator output. Connect to VCORE with 3.3 μ F filter capacitor.
21,41,45	GND	POWER Input	0V Ground input.

Table 3.14 Power and Ground for 56-pin VQFN package

Pin No.	Name	Type	Description
3	OSCI	INPUT	Oscillator input.
4	OSCO	OUTPUT	Oscillator output.
6	REF	INPUT	Current reference – connect via a 12 K Ω resistor @ 1% to GND.
7	DM	I/O	USB Data Signal Minus.
8	DP	I/O	USB Data Signal Plus.
10	TEST	INPUT	IC test pin – for normal operation should be connected to GND.
11	RESET#	INPUT	Reset input (active low).
54	PWREN#	OUTPUT	Active low power-enable output. PWREN# = 0: Normal operation. PWREN# = 1: USB SUSPEND mode or device has not been configured. This can be used by external circuitry to power down logic when device is in USB suspend or has not been configured.
30	SUSPEND#	OUTPUT	Active low when USB is in suspend mode.
53	PWRSV#	INPUT	USB Power Save input. This is an EEPROM configurable option used when the FT2232H is used in a self-powered mode and is used to prevent forcing current down the USB lines when the host or hub is powered off. PWRSV# = 1 : Normal Operation PWRSV# = 0: FT2232H forced into SUSPEND mode. PWRSV# can be connected to GND (via a 10 K Ω resistor) and another resistor (e.g. 4K7) connected to the VBUS of the USB connector. When this input goes high, then it indicates to the FT2232H that it is connected to a host PC. When the host or hub is powered down then the FT2232H is held in SUSPEND mode.

Table 3.15 Common Function pins for 56-pin VQFN package

Pin No.	Name	Type	Description
1	EECS	I/O	EEPROM – Chip Select. Tri-State during device reset.
56	EECLK	OUTPUT	Clock signal to EEPROM. Tri-State during device reset. When not in reset, this outputs the EEPROM clock.
55	EEDATA	I/O	EEPROM – Data I/O Connect directly to Data-In of the EEPROM and to Data-Out of the EEPROM via a 2.2 K resistor. Also, pull Data-Out of the EEPROM to VCC via a 10 K resistor for correct operation. Tri-State during device reset.

Table 3.16 EEPROM Interface Group for 56-pin VQFN package

3.2.4 Configured Pins for FT2232H-56Q

The following sections describe the function of the configurable pins referred to Table 3.13 which is determined by how the FT2232H is configured.

3.2.4.1 FT2232H-56Q pins used in an RS232 interface

The FT2232H-56Q channel A or channel B can be configured as an RS232 interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.17.

Channel A Pin No.	Channel B Pin No.	Name	Type	RS232 Configuration Description
12	32	TXD	OUTPUT	TXD = transmitter output
13	33	RXD	INPUT	RXD = receiver input
14	34	RTS#	OUTPUT	RTS# = Ready To send handshake output
15	35	CTS#	INPUT	CTS# = Clear To Send handshake input
17	37	DTR#	OUTPUT	DTR# = Data Transmit Ready modem signaling line
18	38	DSR#	INPUT	DSR# = Data Set Ready modem signaling line
19	39	DCD#	INPUT	DCD# = Data Carrier Detect modem signaling line
20	40	RI#	INPUT	RI# = Ring Indicator Control Input. When the Remote Wake up option is enabled in the EEPROM, taking RI# low can be used to resume the PC USB Host controller from suspend. (Also see Note 1, 2, 3 in section 4.12)
22	42	TXDEN	OUTPUT	TXDEN = (TTL level). For use with RS485 level converters.
25	48	RXLED#	OUTPUT	RXLED = Receive signaling output when data is transferred from FT2232H to USB Host. Pulses low when receiving data (RXD) via USB. This should be connected to an LED.
26	49	TXLED#	OUTPUT	TXLED = Transmit signaling output when data is transferred from USB Host to FT2232H. Pulses low when transmitting data (TXD) via USB. This should be connected to an LED.

Table 3.17 Channel A and Channel B RS232 Configured Pin Descriptions for FT2232H-56Q

3.2.4.2 FT2232H-56Q pins used in an FT245 Style Synchronous FIFO Interface

The FT2232H-56Q only channel A can be configured as a FT245 style synchronous FIFO interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.18. To enter this mode the external EEPROM must be set to make port A 245 mode. A software command (Set Bit Mode option) is then sent by the application to the FTDI driver to tell the chip to enter single channel synchronous FIFO mode. In this mode the 'B' channel is not available as all resources have been switched onto channel A. In this mode, data is written or read on the rising edge of the CLKOUT.

Channel A Pin No.	Name	Type	FT245 Configuration Description
20,19,18,17 15,14,13,12	ADBUS[7:0]	I/O	D7 to D0 bidirectional FIFO data. This bus is normally input unless OE# is low.
22	RXF#	OUTPUT	When high, do not read data from the FIFO. When low, there is

Channel A Pin No.	Name	Type	FT245 Configuration Description
			data available in the FIFO which can be read by driving RD# low. When in synchronous mode, data is transferred on every clock that RXF# and RD# are both low. Note that the OE# pin must be driven low at least 1 clock period before asserting RD# low.
23	TXE#	OUTPUT	When high, do not write data into the FIFO. When low, data can be written into the FIFO by driving WR# low. When in synchronous mode, data is transferred on every clock that TXE# and WR# are both low.
24	RD#	INPUT	Enables the current FIFO data byte to be driven onto D0...D7 when RD# goes low. The next FIFO data byte (if available) is fetched from the receive FIFO buffer each CLKOUT cycle until RD# goes high.
25	WR#	INPUT	Enables the data byte on the D0...D7 pins to be written into the transmit FIFO buffer when WR# is low. The next FIFO data byte is written to the transmit FIFO buffer each CLKOUT cycle until WR# goes high.
27	CLKOUT	OUTPUT	60 MHz Clock driven from the chip. All signals should be synchronized to this clock.
28	OE#	INPUT	Output enable when low to drive data onto D0-7. This should be driven low at least 1 clock period before driving RD# low to allow for data buffer turn-around.
26	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# = 0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.18 Channel A FT245 Style Synchronous FIFO Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.4.

3.2.4.3 FT2232H-56Q pins used in an FT245 Style Asynchronous FIFO Interface

The FT223-56Q channel A or channel B can be configured as a FT245 asynchronous FIFO interface. When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.19. To enter this mode the external EEPROM must be set to make port A or B or both 245 mode. In this mode, data is written or read on the falling edge of the RD# or WR# signals.

Channel A Pin No.	Channel B Pin No.	Name	Type	FT245 Configuration Description
20,19,18,17,15,14,13,12	40,39,38,37,35,34,33,32	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional FIFO data. This bus is normally input unless RD# is low.
22	42	RXF#	OUTPUT	When high, do not read data from the FIFO. When low, there is data available in the FIFO which can be read by driving RD# low. When RD# goes high again RXF# will always go high and only become low again if there is another byte to read. During reset this signal pin is tri-state, but pulled up to VCCIO via an internal 200 KΩ resistor.
23	46	TXE#	OUTPUT	When high, do not write data into the FIFO.

Channel A Pin No.	Channel B Pin No.	Name	Type	FT245 Configuration Description
				When low, data can be written into the FIFO by strobing WR# high, then low. During reset this signal pin is tri-state, but pulled up to VCCIO via an internal 200 K Ω resistor.
24	47	RD#	INPUT	Enables the current FIFO data byte to be driven onto D0...D7 when RD# goes low. Fetches the next FIFO data byte (if available) from the receive FIFO buffer when RD# goes high.
25	48	WR#	INPUT	Writes the data byte on the D0...D7 pins into the transmit FIFO buffer when WR# goes from high to low.
26	49	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# = 0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.19 Channel A and Channel B FT245 Style Asynchronous FIFO Configured Pin Descriptions for FT2232H-56Q

3.2.4.4 FT2232H-56Q pins used in a Synchronous or Asynchronous Bit-Bang Interface

The FT2232H-56Q channel A or channel B can be configured as a synchronous or asynchronous bit-bang interface. Bit-bang mode is a special FTDI FT2232H device mode that changes the 8 IO lines on either (or both) channels into an 8 bit bi-directional data bus. There are two types of bit-bang modes: synchronous and asynchronous. When configured in any bit-bang mode, the pins used and the descriptions of the signals are shown in Table 3.20.

Channel A Pin No.	Channel B Pin No.	Name	Type	Configuration Description
20,19,18,17,15,14,13,12	40,39,38,37,35,34,33,32	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional Bit-Bang parallel I/O data pins
23	46	WRSTB#	OUTPUT	Write strobe, active low output indicates when new data has been written to the I/O pins from the Host PC (via the USB interface).
24	47	RDSTB#	OUTPUT	Read strobe, this output rising edge indicates when data has been read from the parallel I/O pins and sent to the Host PC (via the USB interface).
26	49	SIWU	INPUT	The Send Immediate / WakeUp signal combines two functions on a single pin. If USB is in suspend mode (PWREN# = 1) and remote wakeup is enabled in the EEPROM, strobing this pin low will cause the device to request a resume on the USB Bus. Normally, this can be used to wake up the Host PC. During normal operation (PWREN# =

Channel A Pin No.	Channel B Pin No.	Name	Type	Configuration Description
				0), if this pin is strobed low any data in the device TX buffer will be sent out over USB on the next Bulk-IN request from the drivers regardless of the pending packet size. This can be used to optimize USB transfer speed for some applications. Tie this pin to VCCIO if not used. (Also see note 1, 2, 3 in section 4.12)

Table 3.20 Channel A and Channel B Synchronous or Asynchronous Bit-Bang Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.10.

3.2.4.5 FT2232H-56Q pins used in an MPSSE

The FT2232H-56Q channel A and channel B, each have a Multi-Protocol Synchronous Serial Engine (MPSSE). Each MPSSE can be independently configured to a number of industry standard serial interface protocols such as JTAG, I2C or SPI, or it can be used to implement a proprietary bus protocol. For example, it is possible to use one of the FT2232H's channels to connect to an SRAM configurable FPGA such as supplied by Altera or Xilinx. The FPGA device would normally be un-configured (i.e. have no defined function) at power-up. Application software on the PC could use the MPSSE to download configuration data to the FPGA over USB. This data would define the hardware function on power up. The other FT2232H channel would be available for another function. Alternatively each MPSSE can be used to control a number of GPIO pins. When configured in this mode, the pins used and the descriptions of the signals are shown Table 3.21.

Channel A Pin No.	Channel B Pin No.	Name	Type	MPSSE Configuration Description
12	32	TCK/SK	OUTPUT	Clock Signal Output. For example: JTAG – TCK, Test interface clock SPI – SK, Serial Clock
13	33	TDI/DO	OUTPUT	Serial Data Output. For example: JTAG – TDI, Test Data Input SPI – DO
14	34	TDO/DI	INPUT	Serial Data Input. For example: JTAG – TDO, Test Data output SPI – DI, Serial Data Input
15	35	TMS/CS	OUTPUT	Output Signal Select. For example: JTAG – TMS, Test Mode Select SPI – CS, Serial Chip Select
17	37	GPIOL0	I/O	General Purpose input/output
18	38	GPIOL1	I/O	General Purpose input/output
19	39	GPIOL2	I/O	General Purpose input/output
20	40	GPIOL3	I/O	General Purpose input/output
22	42	GPIOH0	I/O	General Purpose input/output
23	46	GPIOH1	I/O	General Purpose input/output
24	47	GPIOH2	I/O	General Purpose input/output
25	48	GPIOH3	I/O	General Purpose input/output
26	49	GPIOH4	I/O	General Purpose input/output
27	51	GPIOH5	I/O	General Purpose input/output
28	52	GPIOH6	I/O	General Purpose input/output
29	53	GPIOH7	I/O	General Purpose input/output

Table 3.21 Channel A and Channel B MPSSE Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.6.

3.2.4.6 FT2232H-56Q Pins used as a Fast Serial Interface

The FT2232H-56Q channel B can be configured for use with high-speed optical bi-directional isolated serial data transfer: Fast Serial Interface. (Not available on channel A). A proprietary FTDI protocol designed to allow galvanic isolated devices to communicate synchronously with the FT2232H-56Q using just 4 signal wires (over two dual opto-isolators), and two power lines. The peripheral circuitry controls the data transfer rate in both directions, whilst maintaining full data integrity. Maximum USB full speed data rates can be achieved. Both 'A' and 'B' channels can communicate over the same 4 wire interface if desired.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.22.

Channel B Pin No.	Name	Type	Fast Serial Interface Configuration Description
32	FSDI	INPUT	Fast serial data input.
33	FSCLK	INPUT	Fast serial clock input. Clock input to FT2232H chip to clock data in or out.
34	FSDO	OUTPUT	Fast serial data output.
35	FSCTS	OUTPUT	Fast serial Clear To Send signal output. Driven low to indicate that the chip is ready to send data

Table 3.22 Channel B Fast Serial Interface Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.8.

3.2.4.7 FT2232H-56Q Pins Configured as a CPU-style FIFO Interface

The FT2232H-56Q channel A or channel B can be configured in a CPU-style FIFO interface mode which allows a CPU to interface to USB via the FT2232H-56Q. This mode is enabled in the external EEPROM.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.23.

Channel A Pin No.	Channel B Pin No.	Name	Type	Fast Serial Interface Configuration Description
20,19,18,17 15,14,13,12	40,39,38,37, 35,34,33,32	Channel A = ADBUS[7:0] Channel B = BDBUS[7:0]	I/O	D7 to D0 bidirectional data bus
22	42	CS#	INPUT	Active low chip select input
23	46	A0	INPUT	Address bit A0
24	47	RD#	INPUT	Active Low FIFO Read input
25	48	WR#	INPUT	Active Low FIFO Write input

Table 3.23 Channel A and Channel B CPU-style FIFO Interface Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.9.

3.2.4.8 FT2232H-56Q Pins Configured as a Host Bus Emulation Interface

The FT2232H-56Q can be used to combine channel A and channel B to be configured as a host bus emulation interface mode which emulates a standard 8048 or 8051 MCU host.

When configured in this mode, the pins used and the descriptions of the signals are shown in Table 3.24.

Pin No.	Name	Type	Fast Serial Interface Configuration Description
20,19,18,17 15,14,13,12	ADBUS[7:0]	I/O	Multiplexed bidirectional Address/Data bus AD7 to AD0
29,28,27,26 25,24,23,22	A[15:8]	OUTPUT	Extended Address A15 to A8
32	CS#	OUTPUT	Active low chip select device during Read or Write.
33	ALE	OUTPUT	Positive pulse to latch the address
34	RD#	OUTPUT	Active low read output.
35	WR#	OUTPUT	Active low write output. (Data is setup before WR# goes low, and is held after WR# goes high)
37	IORDY	INPUT	Extends the time taken to perform a Read or Write operation if driven low. Pull up to V _{CORE} if not being used.
38	CLKOUT	OUTPUT	Master clock. Outputs the clock signal being used by the configured interface.
39	I/O0	I/O	MPSSE mode instructions to set / clear or read the high byte of data can be used with this pin. Please refer to application note AN_108 Command Processor For MPSSE and MCU Host Bus Emulation Modes for operation of these instructions.
40	I/O1	I/O	MPSSE mode instructions to set / clear or read the high byte of data can be used with this pin. In addition this pin has instructions which will make the controller wait until it is high, or wait until it is low. This can be used to connect to an IRQ pin of a peripheral chip. The FT2232H will wait for the interrupt, and then read the device, and pass the answer back to the host PC. I/O1 must be held in input mode if this option is used. Please refer to application note AN_108 Command Processor For MPSSE and MCU Host Bus Emulation Modes for operation of these instructions.

Table 3.24 Channel A and Channel B Host Bus Emulation Interface Configured Pin Descriptions for FT2232H-56Q

For a functional description of this mode, please refer to Section 4.7.

4 Function Description

The FT2232H USB 2.0 High Speed (480 Mb/s) to UART/FIFO is one of FTDI's 5th generation of ICs. It has the capability of being configured in a variety of industry standard serial or parallel interfaces.

The FT2232H has two independent configurable interfaces. Each interface can be configured as UART, FIFO, JTAG, SPI, I2C or bit-bang mode with independent baud rate generators. In addition to these, the FT2232H supports a host bus emulation mode, a CPU-Style FIFO mode and a fast opto-isolated serial interface mode.

4.1 Key Features

USB High Speed to Dual Interface. The FT2232H is a USB 2.0 High Speed (480 Mbits/s) to dual independent flexible and configurable parallel/serial interfaces.

Functional Integration. The FT2232H integrates a USB protocol engine which controls the physical Universal Transceiver Macrocell Interface (UTMI) and handles all aspects of the USB 2.0 High Speed interface. The FT222H includes an integrated +1.8 V Low Drop-Out (LDO) regulator and 12 MHz to 480 MHz PLL. It also includes 4 kbytes Tx and Rx data buffers per interface. The FT2232H effectively integrates the entire USB protocol on a chip with no firmware required.

MPSSE. Multi-Purpose Synchronous Serial Engines (MPSSE), capable of speeds up to 30 Mbits/s, provides flexible synchronous interface configurations.

Data Transfer rate. The FT2232H supports a data transfer rate up to 12 Mbaud when configured as an RS232/RS422/RS485 UART interface or greater than 25 Mbytes/second over a synchronous parallel FIFO interface. Please note the FT2232H does not support the baud rates of 7 Mbaud 9 Mbaud, 10 Mbaud and 11 Mbaud.

Latency Timer. This is really a feature of the driver and is used to as a timeout to flush short packets of data back to the PC. The default is 16 ms, but it can be altered between 0ms and 255 ms. At 0 ms latency you get a packet transfer on every high speed microframe.

4.2 Functional Block Descriptions

Dual Multi-Purpose UART/FIFO Controllers. The FT2232H has two independent UART/FIFO Controllers. These control the UART data, 245 FIFO data, opto isolation (Fast Serial) or control the Bit-Bang mode if selected by SETUP command. Each Multi-Purpose UART/FIFO Controller also contains an MPSSE (Multi-Protocol Synchronous Serial Engine) which can be used independently of each other. Using this MPSSE, the Multi-Purpose UART/FIFO Controller can be configured, under software command, to have 1 MPSSE + 1 UART / 245 FIFO (each UART / 245 can be set to Bit Bang mode to gain extra I/O if required) or 2 MPSSE.

USB Protocol Engine and FIFO control. The USB Protocol Engine controls and manages the interface between the UTMI PHY and the FIFOs of the chip. It also handles power management and the USB protocol specification.

Dual Port FIFO TX Buffer (4 Kbytes per interface). Data from the Host PC is stored in these buffers to be used by the Multi-purpose UART/FIFO controllers. This is controlled by the USB Protocol Engine and FIFO control block.

Dual Port FIFO RX Buffer (4 Kbytes per interface). Data from the Multi-purpose UART/FIFO controllers is stored in these blocks to be sent back to the Host PC when requested. This is controlled by the USB Protocol Engine and FIFO control block.

RESET Generator. The integrated Reset Generator Cell provides a reliable power-on reset to the device internal circuitry at power up. The RESET# input pin allows an external device to reset the FT2232H. RESET# should be tied to VCCIO (+3.3 V) if not being used.

Independent Baud Rate Generators. The Baud Rate Generators provides an x16 or an x10 clock input to the UART's from a 120 MHz reference clock and consists of a 14 bit pre-scaler and 4 register bits which provide fine tuning of the baud rate (used to divide by a number plus a fraction). This determines the Baud Rate of the UART which is programmable from 183 baud to 12 million baud. The FT2232H does not

support the baud rates of 7 Mbaud 9 Mbaud, 10 Mbaud, and 11 Mbaud.

See application note [AN232B-05 Configuring FT232R, FT2232 and FT232BM Baud Rates](#) for more details.

+1.8 V LDO Regulator. The +1.8 V LDO regulator generates the +1.8 V for the core and the USB transceiver cell. Its input (VREGIN) must be connected to a +3.3 V external power source. It is also recommended to add an external filtering capacitor to the VREGIN. There is no direct connection from the +1.8 V output (VREGOUT) and the internal functions of the FT2232H. The PCB must be routed to connect VREGOUT to the pins that require the +1.8 V including VCORE.

UTMI PHY. The Universal Transceiver Macrocell Interface (UTMI) physical interface cell. This block handles the Full speed / High Speed SERDES (serialise – deserialise) function for the USB TX/RX data. It also provides the clocks for the rest of the chip. A 12 MHz crystal should be connected to the OSCI and OSCO pins. A 12 K Ω resistor should be connected between REF and GND on the PCB.

The UTMI PHY functions include:

- Supports 480 Mbit/s “High Speed” (HS)/ 12 Mbit/s “Full Speed” (FS), FS Only and “Low Speed” (LS)
- SYNC/EOP generation and checking
- Data and clock recovery from serial stream on the USB.
- Bit-stuffing/unstuffing; bit stuff error detection.
- Manages USB Resume, Wake Up and Suspend functions.
- Single parallel data clock output with on-chip PLL to generate higher speed serial data clocks.

EEPROM Interface. When used without an external EEPROM the FT2232H defaults to a USB to dual asynchronous serial port device. Adding an external 93C46 (93C56 or 93C66) EEPROM allows each of the chip’s channels to be independently configured as a serial UART (RS232 mode), parallel FIFO (245) mode or fast serial (opto isolation). The external EEPROM can also be used to customise the USB VID, PID, Serial Number, Product Description Strings and Power Descriptor value of the FT2232H for OEM applications. Other parameters controlled by the EEPROM include Remote Wake Up, Soft Pull Down on Power-Off and I/O pin drive strength.

The EEPROM should be a 16 bit wide configuration such as a Microchip 93LC46B or equivalent capable of a 1 Mbit/s clock rate at VCC = +3.00 V to 3.6 V. The EEPROM is programmable in-circuit over USB using a utility program called [FT_PROG](#). This allows a blank part to be soldered onto the PCB and programmed as part of the manufacturing and test process.

If no EEPROM is connected (or the EEPROM is blank), the FT2232H will default to dual serial ports. The device uses its built-in default VID (0403), PID (6010) Product Description and Power Descriptor Value. In this case, the device will not have a serial number as part of the USB descriptor.

4.3 Dual Port FT232 UART Interface Mode Description

The FT2232H can be configured in similar UART modes as the FTDI FT232 devices. The following examples illustrate how to configure the FT2232H with an RS232, RS422 or RS485 interface. The FT2232 can be configured as a mixture of these interfaces.

4.3.1 Dual Port RS232 Configuration

Figure 4.1 illustrates how the FT2232H can be configured with an RS232 UART interface. This can be repeated for channel B to provide a dual RS232, but has been omitted for clarity.

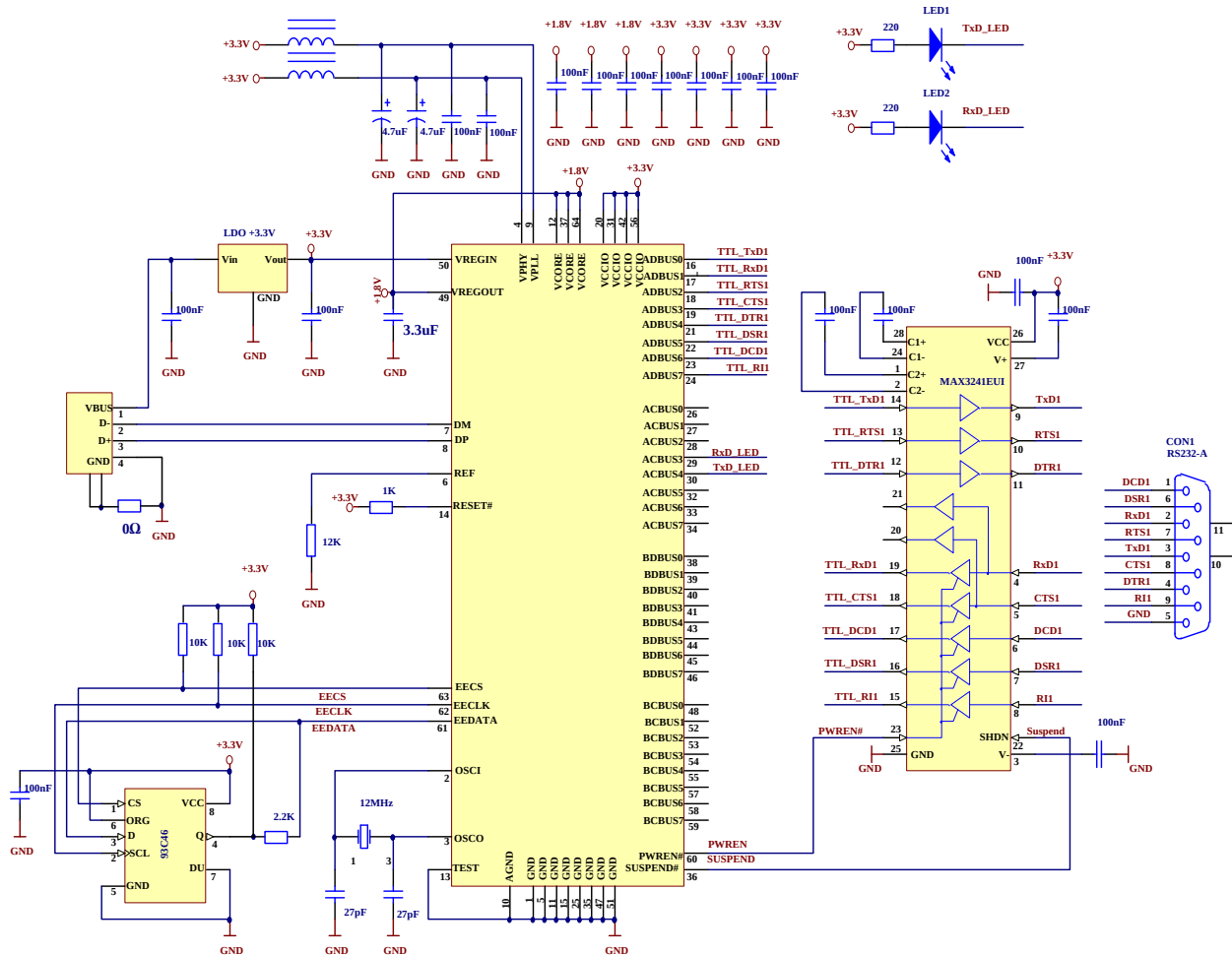


Figure 4.1 RS232 Configuration

4.3.2 Dual Port RS422 Configuration

Figure 4.2 illustrates how the FT2232H can be configured as a dual RS422 interface.

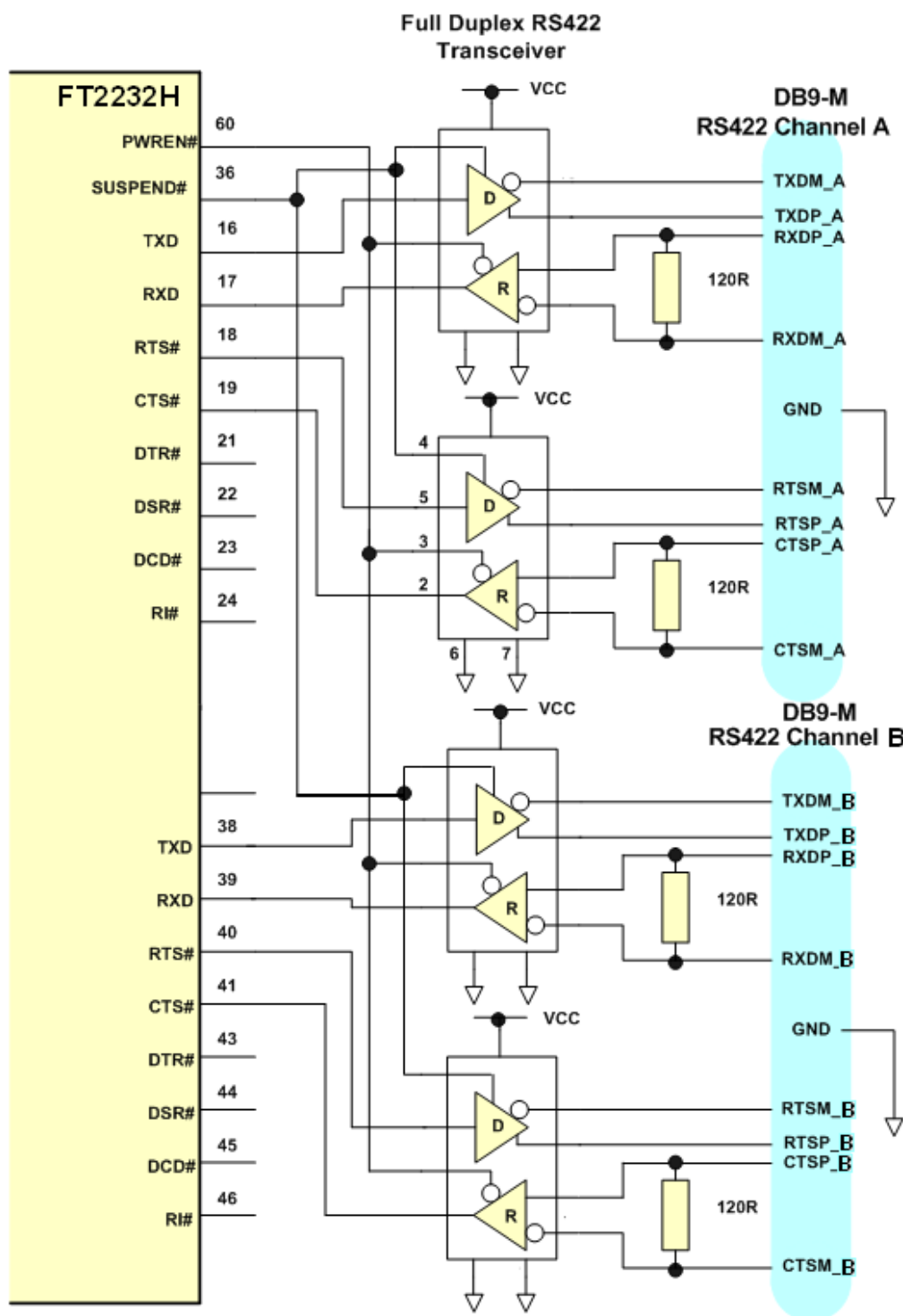


Figure 4.2 Dual RS422 Configuration

In this case both channel A and channel B are configured as UART operating at TTL levels and a level converter device (full duplex RS485 transceiver) is used to convert the TTL level signals from the FT2232H to RS422 levels. The PWREN# signal is used to power down the level shifters such that they operate in a low quiescent current when the USB interface is in suspend mode.

4.3.3 Dual Port RS485 Configuration

Figure 4.3 illustrates how the FT2232H can be configured as a dual RS485 interface.

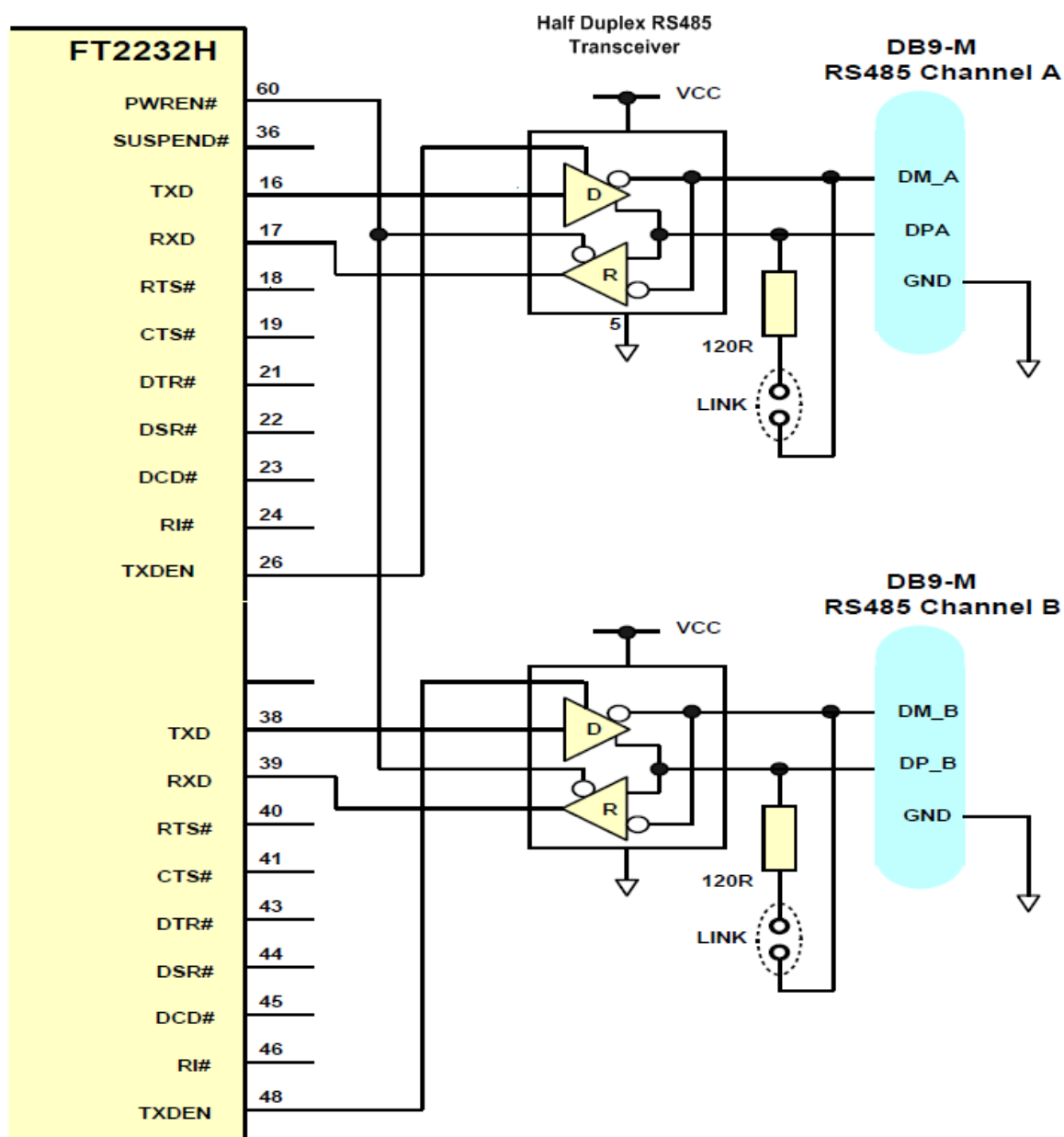


Figure 4.3 Dual RS485 Configuration

In this case both channel A and channel B are configured as RS485 operating at TTL levels and a level converter device (half duplex RS485 transceiver) is used to convert the TTL level signals from the FT232H to RS485 levels. It has separate enables on both the transmitter and receiver. With RS485, the transmitter is only enabled when a character is being transmitted from the UART. The TXDEN pins on the FT2232H are provided for exactly that purpose, and so the transmitter enables are wired to the TXDEN's. The receiver enable is active low, so it is wired to the PWREN# pin to disable the receiver when in USB suspend mode.

RS485 is a multi-drop network – i.e. many devices can communicate with each other over a single two wire cable connection. The RS485 cable requires to be terminated at each end of the cable. Links are provided to allow the cable to be terminated if the device is physically positioned at either end of the cable.

In this example the data transmitted by the FT2232H is also received by the device that is transmitting. This is a common feature of RS485 and requires the application software to remove the transmitted data from the received data stream. With the FT2232H it is possible to do this entirely in hardware – simply modify the schematic so that RXD of the FT2232H is the logical OR of the level converter device receiver output with TXDEN using an HC32 or similar logic gate.

4.4 FT245 Synchronous FIFO Interface Mode Description

When channel A is configured in an FT245 Synchronous FIFO interface mode the IO timing of the signals used are shown in

Figure 4.4, which shows details for read and write accesses. The timings are shown in Table 4.1.

Note that only a read or a write cycle can be performed at any one time. Data is read or written on the rising edge of the CLKOUT clock.

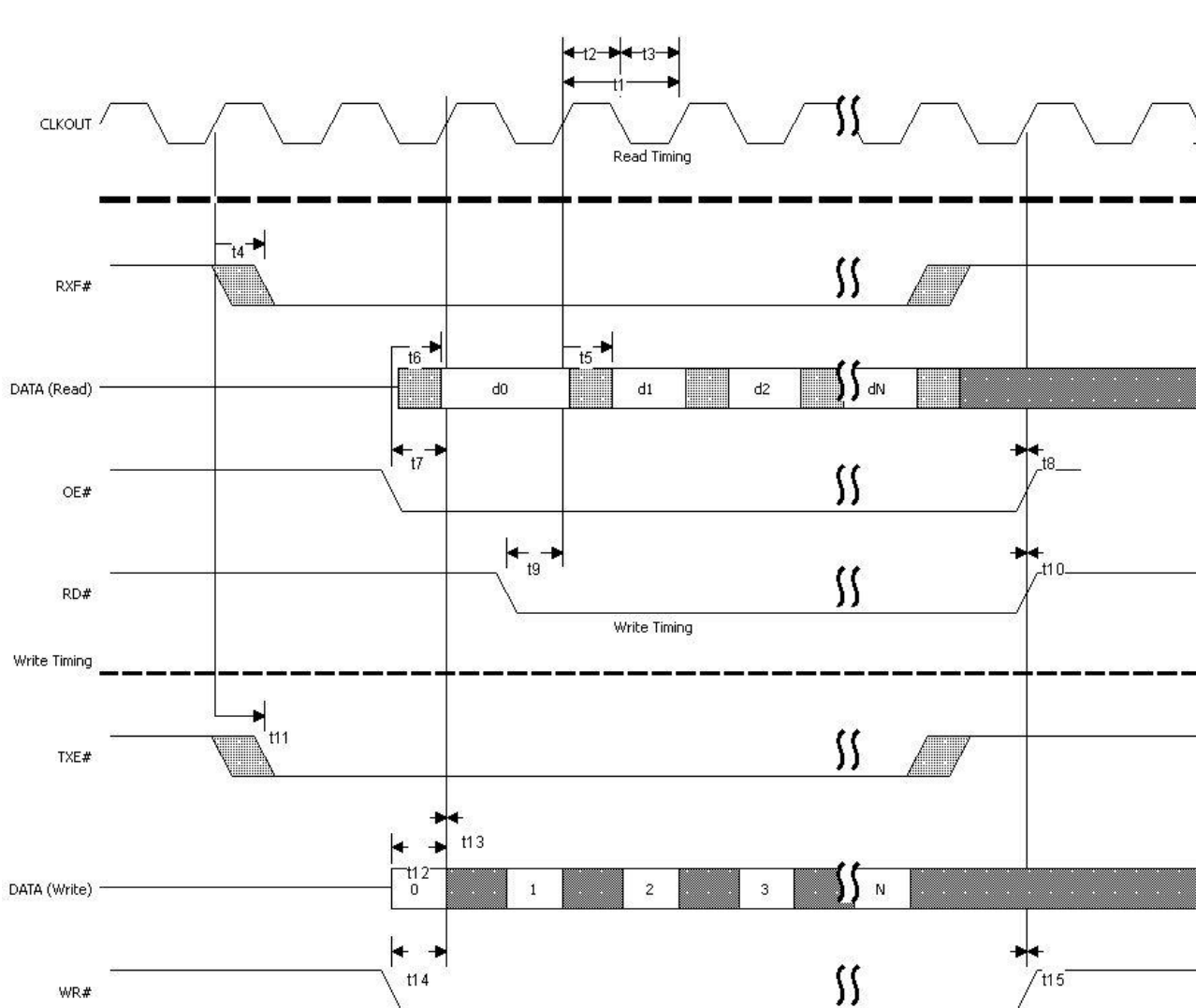


Figure 4.4 FT245 Synchronous FIFO Interface Signal Waveforms

Name	Minimum	Typical	Maximum	Units	Description
t1		16.67	16.67	ns	CLKOUT period

Name	Minimum	Typical	Maximum	Units	Description
t2	7.5	8.33	9.17	ns	CLKOUT high period
t3	7.5	8.33	9.17	ns	CLKOUT low period
t4	1		7.15	ns	CLKOUT to RXF#
t5	1		7.15	ns	CLKOUT to read DATA valid
t6	1		7.15	ns	OE# to read DATA valid
t7	8		16.67	ns	OE# setup time
t8	0			ns	OE# hold time
t9	8		16.67	ns	RD# setup time to CLKOUT (RD# low after OE# low)
t10	0			ns	RD# hold time
t11	1		7.15	ns	CLKOUT TO TXE#
t12	8		16.67	ns	Write DATA setup time
t13	0			ns	Write DATA hold time
t14	8		16.67	ns	WR# setup time to CLKOUT (WR# low after TXE# low)
t15	0			ns	WR# hold time

Table 4.1 FT245 Synchronous FIFO Interface Signal Timings

This single channel mode uses a synchronous interface to get high data transfer speeds. The chip drives a 60 MHz CLKOUT clock for the external system to use.

Note that Asynchronous FIFO mode must be selected on both channels before selecting the Synchronous FIFO mode in software.

4.4.1 FT245 Synchronous FIFO Read Operation

A read operation is started when the chip drives RXF# low. The external system can then drive OE# low to turn around the data bus drivers before acknowledging the data with the RD# signal going low. The first data byte is on the bus after OE# is low. The external system can burst the data out of the chip by keeping RD# low or it can insert wait states in the RD# signal. If there is more data to be read it will change on the clock following RD# sampled low. Once all the data has been consumed, the chip will drive RXF# high. Any data that appears on the data bus, after RXF# is high, is invalid and should be ignored.

4.4.2 FT245 Synchronous FIFO Write Operation

A write operation can be started when TXE# is low. WR# is brought low when the data is valid. A burst operation can be done on every clock providing TXE# is still low. The external system must monitor TXE# and its own WR# to check that data has been accepted. Both TXE# and WR# must be low for data to be accepted.

4.5 FT245 Asynchronous FIFO Interface Mode Description

The FT2232H can be configured as a dual channel asynchronous FIFO interface. This mode is similar to the synchronous FIFO interface with the exception that the data is written to or read from the FIFO on the falling edge of the WR# or RD# signals.

This mode does not provide a CLKOUT signal and it does not expect an OE# input signal. The following diagrams illustrate the asynchronous FIFO mode timing.

Read Timing

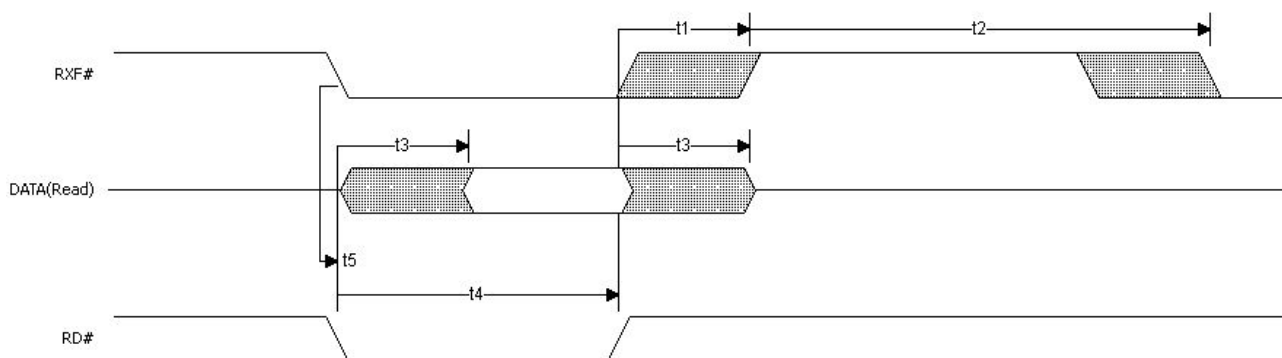


Figure 4.5 FT245 asynchronous FIFO Interface READ Signal Waveforms

Write Timing

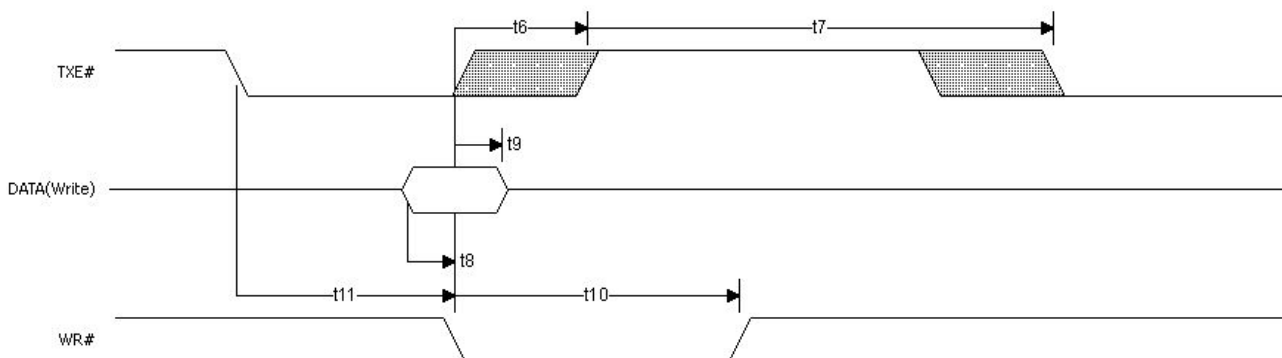


Figure 4.6 FT245 asynchronous FIFO Interface WRITE Signal Waveforms

Name	Minimum	Typical	Maximum	Units	Description
t1	1		14	ns	RD# inactive to RX#
t2	49			ns	RXF# inactive after RD# cycle
t3	1		14	ns	RD# to DATA
t4	30			ns	RD# active pulse width
t5	0			ns	RD# active after RXF#
t6	1		14	ns	WR# active to TXE# inactive
t7	49			ns	TXE# inactive after WR# cycle
t8	5			ns	DATA to WR# active setup time
t9	5			ns	DATA hold time after WR# goes active
t10	30			ns	WR# active pulse width
t11	0			ns	WR# active after TXE#

Table 4.2 Asynchronous FIFO Timings (based on standard drive level outputs)

4.6 MPSSE Interface Mode Description

MPSSE Mode is designed to allow the FT2232H to interface efficiently with synchronous serial protocols such as JTAG, I2C and SPI Bus. It can also be used to program SRAM based FPGA's over USB. The MPSSE interface is designed to be flexible so that it can be configured to allow any synchronous serial protocol (industry standard or proprietary) to be implemented using the FT2232H. MPSSE is available on channel A and channel B.

MPSSE is fully configurable, and is programmed by sending commands down the data stream. These can be sent individually or more efficiently in packets. MPSSE is capable of a maximum sustained data rate of 30 Mbits/s.

When a channel is configured in MPSSE mode, the IO timing and signals used are shown in Figure 4.7 and Table 4.3 These show timings for CLKOUT=30MHz. CLKOUT can be divided internally to provide a slower clock.

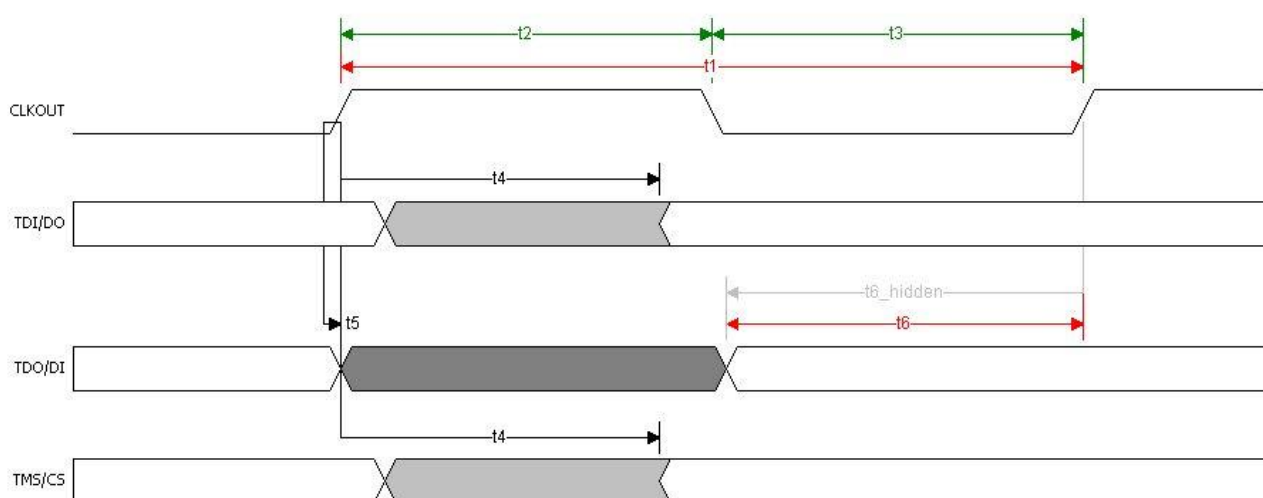


Figure 4.7 MPSSE Signal Waveforms

Name	Minimum	Typical	Maximum	Units	Description
t1		33.33		ns	CLKOUT period
t2	15	16.67		ns	CLKOUT high period
t3	15	16.67		ns	CLKOUT low period
t4	1		7.15	ns	CLKOUT to TDI/DO delay
t5	0			ns	TDO/DI hold time
t6	11			ns	TDO/DI setup time

Table 4.3 MPSSE Signal Timings

MPSSE mode is enabled using Set Bit Bang Mode driver command. A hex value of 2 will enable it, and a hex value of 0 will reset the device. See application note [AN2232-02 Bit Mode Functions for the FT2232](#) for more details and examples.

The MPSSE command set is fully described in application note [AN 108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#).

The following additional application notes are available for configuring the MPSSE:

- [AN 110 Programmers Guide for High-Speed FTCJTAG DLL.pdf](#)
- [AN 177 User Guide for LibMPSSE-I2C](#)
- [AN 178 User Guide for LibMPSSE-SPI](#)

4.6.1 MPSSE Adaptive Clocking

Adaptive clocking is a new MPSSE feature added to the FT2232H MPSSE engine.

The mode is effectively handshaking the CLK signal with a return clock RTCK. This is a technique used by ARM processors.

The FT2232H will assert the CLK line and wait for the RTCK to be returned from the target device to GPIOL3 line before changing the TDO (data out line).

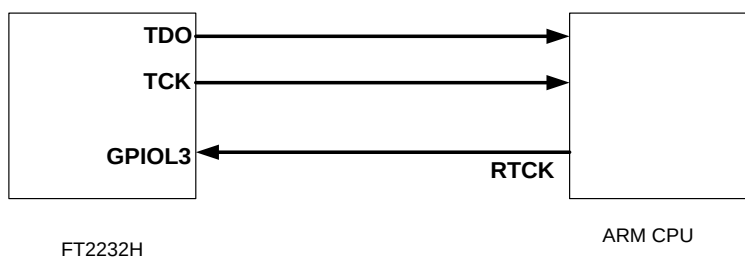


Figure 4.8 Adaptive Clocking Interconnect

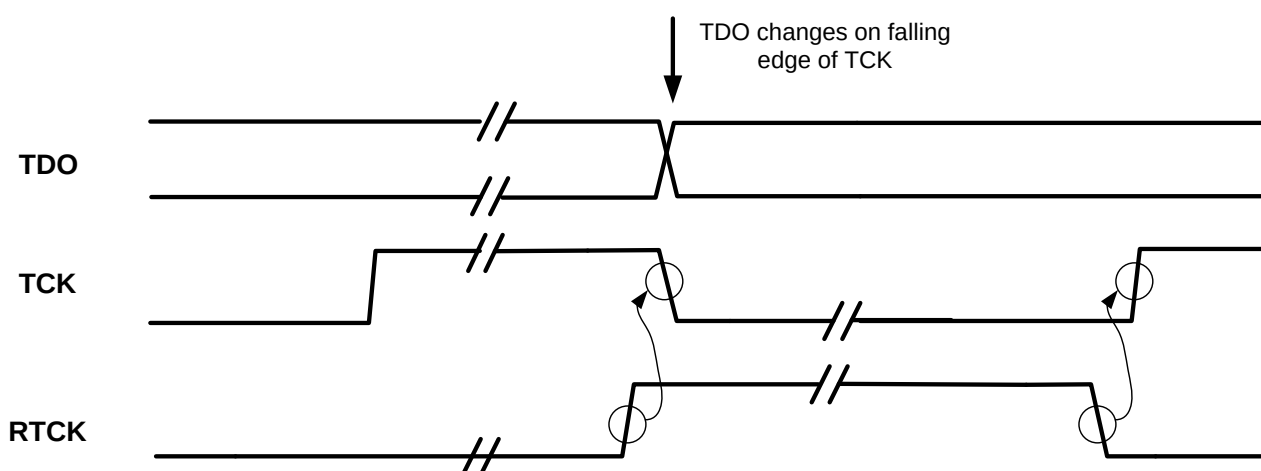


Figure 4.9: Adaptive Clocking waveform

Adaptive clocking is not enabled by default.

See: [AN_108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#)

4.7 MCU Host Bus Emulation Mode

MCU host bus emulation mode uses both of the FT2232H's A and B channel interfaces to make the chip emulate a standard 8048/8051 MCU host bus. This allows peripheral devices for these MCU families to be directly connected to USB via the FT2232H.

The lower 8 bits (AD7 to AD0) is a multiplexed Address / Data bus. A15 to A8 provide upper (extended) addresses. There are 4 basic operations:-

- 1) Read (does not change A15 to A8)
- 2) Read Extended (changes A15 to A8)
- 3) Write (does not change A15 to A8)
- 4) Write Extended (changes A15 to A8)

MCU Host Bus Emulation mode is enabled using Set Bit Bang Mode driver command. A hex value of 8 will enable it, and a hex value of 0 will reset the device. The FT2232H operates in the same way as the FT2232D. See application note [AN2232-02 Bit Mode Functions for the FT2232](#) for more details and examples.

The MCU Host Bus Emulation Mode command set is fully described in application note [AN_108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#).

When MCU Host Bus Emulation mode is enabled the IO signal lines on both channels work together and the pins are configured as described in Table 3.12. The following sections give some details of the read and write cycle waveforms and timings. The CLKOUT output clock can operate up to 60 MHz.

In Host Bus Emulation mode the clock divisor has no effect. The clock divisor is used for serial data and is a different part of the MPSSE block. In host bus emulation the 60MHz clock is always output and does not change with any commands.

4.7.1 MCU Host Bust Emulation Mode Signal Timing – Write Cycle

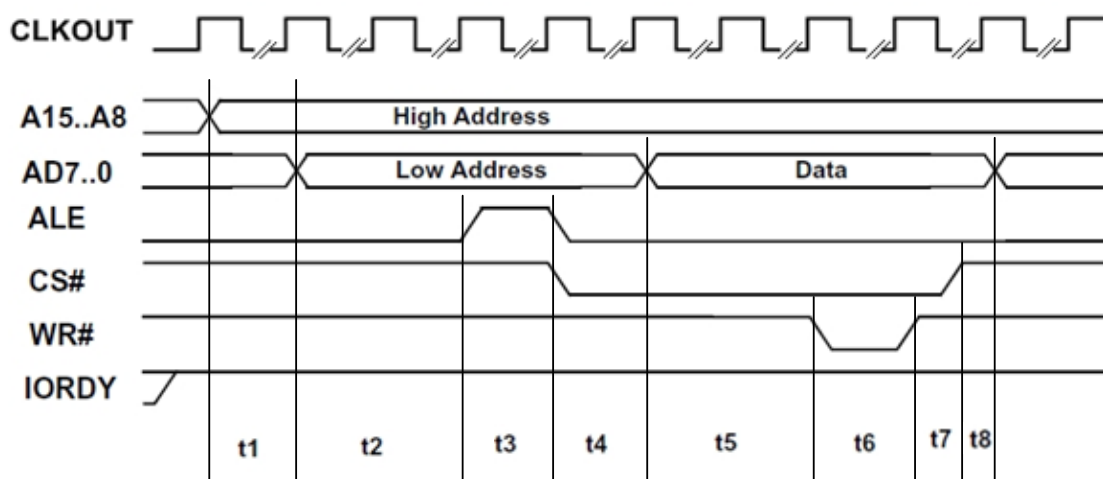


Figure 4.10 MCU Host Bus Emulation Mode Signal Waveforms – write cycle

Div by 5	Number of clock cycles for Write				Description
	On		Off		
IORDY	Low	High	Low	High	
t1	6	6	2	2	A15-8 to AD7-0
t2	3	3	1	1	AD7-0 to ALE
t3	10	5	2	1	ALE width
t4	2	2	2	2	ALE to Address not Valid
t5	13	3	1	1	AD7-0 Data Valid to WRn
t6	10	5	6	1	WRn width
t7	1	1	1	1	WRn inactive to CSn inactive
t8	1	1	1	1	CSn inactive to AD7-0 not valid

Table 4.4 MCU Host Bus Emulation Mode Signal Timings – write cycle

When Div By 5 is on the device will return 2 bytes when doing a read. When it is off the device will return 1 byte when doing a read. The clock period is 16.67 ns so most devices would need the Div By 5 to be set on. IORDY can be held low permanently to extend all cycles.

4.7.2 MCU Host Bust Emulation Mode Signal Timing – Read Cycle

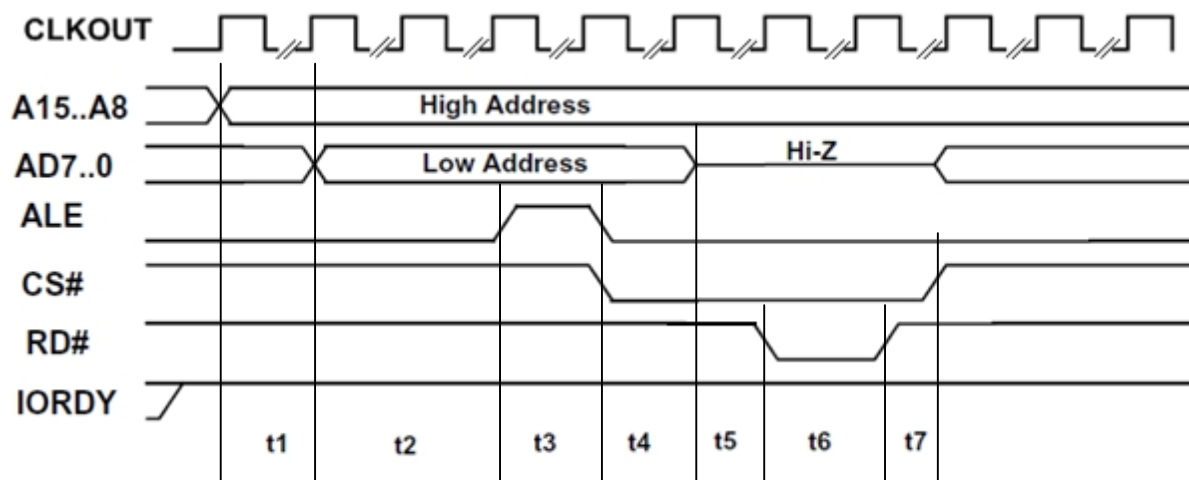


Figure 4.11 MCU Host Bus Emulation Mode Signal Waveforms – read cycle

Div by 5	Number of clock cycles for Read				Description
	On		Off		
IORDY	Low	High	Low	High	
t1	6	6	2	2	A15-8 to AD7-0
t2	3	3	1	1	AD7-0 to ALE
t3	10	5	2	1	ALE width
t4	2	5	2	1	ALE to Address not Valid
t5	13	0	1	0	Address not Valid to RDn active
t6	10	10	6	2	RDn active
t7	1	1	1	1	RDn inactive to CSn inactive

Table 4.5 MCU Host Bus Emulation Mode Signal Timings– read cycle

When Div By 5 is on the device will return 2 bytes when doing a read. When it is off the device will return 1 byte when doing a read. The clock period is 16.67 ns so most devices would need the Div By 5 to be set on. IORDY can be held low permanently to extend all cycles.

An example of the MCU Host Emulation Interface enabling a USB interface to CAN Bus using a CANBus Controller is shown in Figure 4.12.

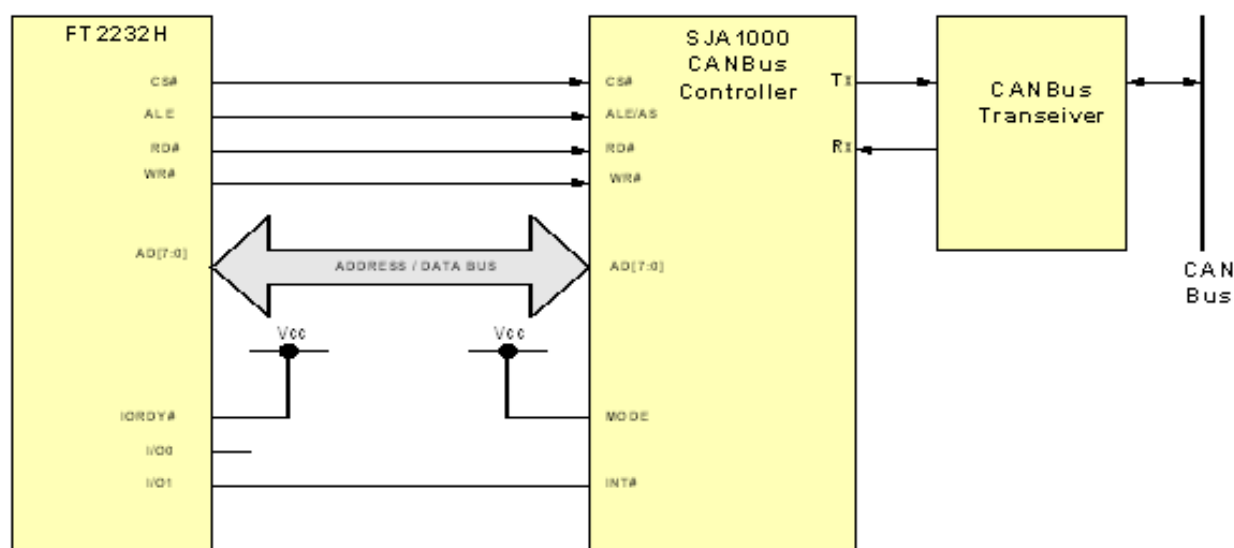


Figure 4.12 MCU Host Emulation Example using a CANBus Controller

4.8 Fast Opto-Isolated Serial Interface Mode Description

Fast Opto-Isolated Serial Interface Mode provides a method of communicating with an external device over USB using 4 wires that can have opto-isolators in their path, thus providing galvanic isolation between systems. If either channel A or channel B is enabled in Fast Opto-Isolated Serial mode then the pins on channel B are switched to the fast serial interface configuration. The I/O interface for fast serial mode is always on channel B, even if both channels are being used in this mode. An address bit is used to determine the source or destination channel of the data. It therefore makes sense to always use at least channel B or both for fast serial mode, but not A own its own.

Fast serial mode is enabled by setting the appropriate bits in the external EEPROM. The fast serial mode can be held in reset by setting a bit value of 10 using the Set Bit Bang Mode command. While this bit is set the device is held reset – data can be sent to the device, but it will not be sent out by the device until the device is enabled again. This is done by sending a bit value of 0 using the set bit mode command. See application note [AN2232-02 Bit Mode Functions for the FT2232](#) for more details and examples.

When either Channel B or both Channel A and B are configured in Fast Opto-Isolated Serial Interface mode the IO timing of the signals used are shown in Figure 4.13 and the timings are shown in Table 4.6.

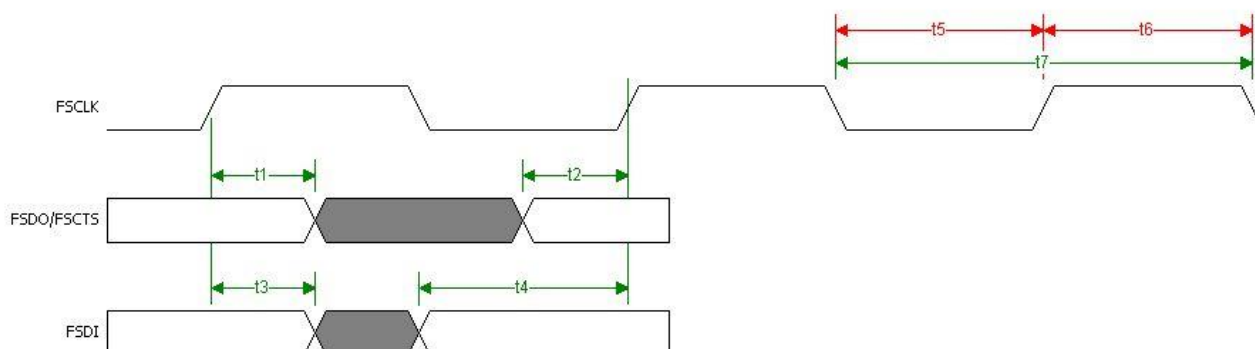


Figure 4.13 Fast Opto-Isolated Serial Interface Signal Waveforms

Name	Minimum	Typical	Maximum	Units	Description
t1	5			ns	FSD O/FSC TS hold time
t2	5			ns	FSD O/FSC TS setup time
t3	5			ns	FSDI hold time
t4	10			ns	FSDI Setup Time
t5	10			ns	FSCLK low
t6	10			ns	FSCLK high
t7	20			ns	FSCLK Period

Table 4.6 Fast Opto-Isolated Serial Interface Signal Timings

4.8.1 Outgoing Fast Serial Data

To send fast serial data out of the FT2232H, the external device must drive the FSCLK clock. If the FT2232H has data ready to send, it will drive FSDO output low to indicate the start bit. It will not do this if it is currently receiving data from the external device. This is illustrated in Figure 4.14.

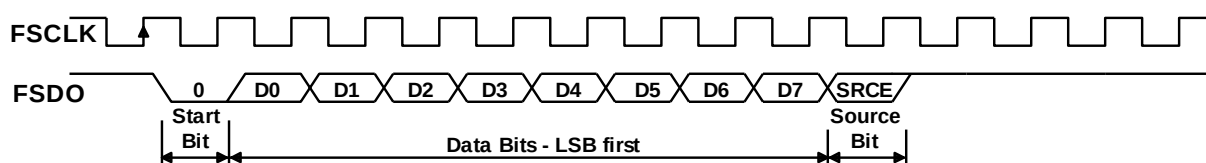


Figure 4.14 Fast Opto-Isolated Serial Interface Output Data

Notes:

1. The first bit output (Start bit) is always 0.
2. FSDO is always sent LSB first.
3. The last serial bit output is the source bit (SRCE). It indicates which channel the data has come from. A '0' means that it has come from Channel A, a '1' means that it has come from Channel B.
4. If the target device is unable to accept the data when it detects the START bit, it should stop the FSCLK until it can accept the data.

4.8.2 Incoming Fast Serial Data

An external device is allowed to send data into the FT2232H if FSCTS is high. On receipt of a zero START bit on FSDI, the FT2232H will drop FSCTS on the next positive clock edge. The data from bits 0 to 7 are then clocked in (LSB first). The last bit (DEST) determines where the data will be written to. The data can be sent to either channel A or to channel B. If DEST= '0', the data is sent to channel A, (assuming channel A is enabled for fast serial mode, otherwise the data is sent to channel B). If DEST= '1' the data is sent to channel B, (assuming channel B is enabled for fast serial mode, otherwise the data will go to channel A. (Either channel A, channel B or both channels must be enabled as fast serial mode or the function is disabled). This is illustrated in Figure 4.15.

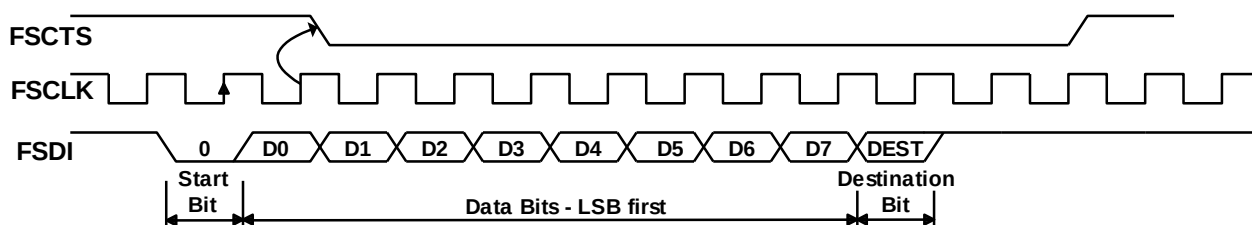


Figure 4.15 Fast Opto-Isolated Serial Interface Input Data

Notes:

1. The first bit input (Start bit) is always 0.
2. FSDI is always received LSB first.
3. The last received serial bit is the destination bit (DEST).It indicates which channel the data should go to. A '0' means that it should go to channel A, a '1' means that it should go to channel B.

4. The target device should ensure that CTS is high before it sends data. CTS goes low after data bit 0 (D0) and stays low until the chip can accept more data.

4.8.3 Fast Opto-Isolated Serial Data Interface Example

The following example, Figure 4.16, shows two Agilent HCPL-2430 (see the semiconductor section at www.agilent.com) high speed opto-couplers used to optically isolate an external device which interfaced to USB using the FT2232H. In this example VCC5V is the USB VBUS supply and VCCE is the supply to the external device.

Care must be taken with the voltage used to power the photo-LED's. It must be the same voltage as that the FT2232H I/Os are driving to, or the LED's may be permanently on. Limiting resistors should be fitted in the lines that drive the diodes. The outputs of the opto-couplers are open-collector and require a pull-up resistor.

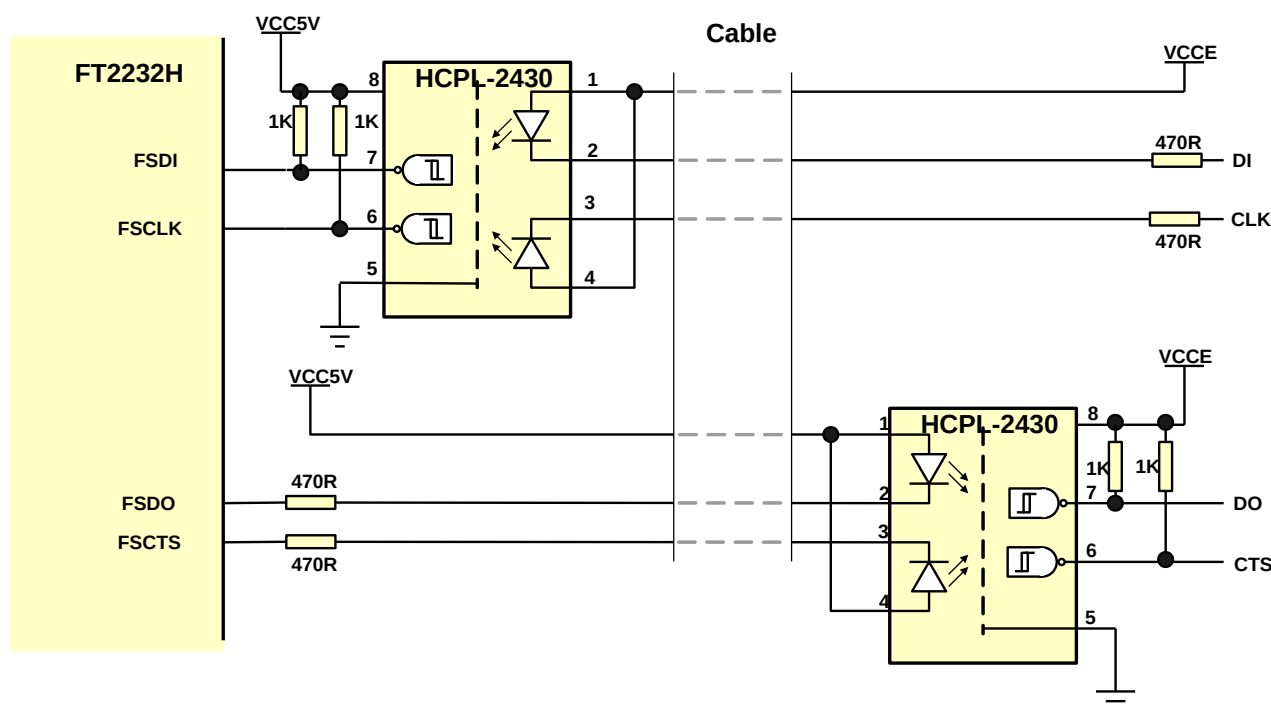


Figure 4.16 Fast Opto-Isolated Serial Interface Example

4.9 CPU-Style FIFO Interface Mode Description

CPU-style FIFO interface mode is designed to allow a CPU to interface to USB via the FT2232H. This mode is enabled in the external EEPROM. The interface is achieved using a chip select bit (CS#) and address bit (A0). When either Channel A or Channel B are in CPU-style Interface mode the IO signal lines are configured as given in Table 3.11.

This mode uses a combination of CS# and A0 to determine the operation to be carried out. The following truth-table, Table 4.7, gives the decode values for particular operations.

CS#	A0	RD#	WR#
1	X	X	X
0	0	Read Data Pipe	Write Data Pipe
0	1	Read Status	Send Immediate

Table 4.7 CPU-Style FIFO Interface Operation Select

The Status read is shown in Table 4.8.

Data Bit	Data	Status
bit 0	1	Data available (=RXF)
bit 1	1	Space available (=TXE)
bit 2	1	Suspend
bit 3	1	Configured
bit 4	X	X
bit 5	X	X
bit 6	X	X
bit 7	X	X

Table 4.8 CPU-Style FIFO Interface Operation Read Status Description

Note that bits 7 to 4 can be arbitrary values and that X= not used.

The timing of reading and writing in this mode is shown in Figure 4.17 and Table 4.9.

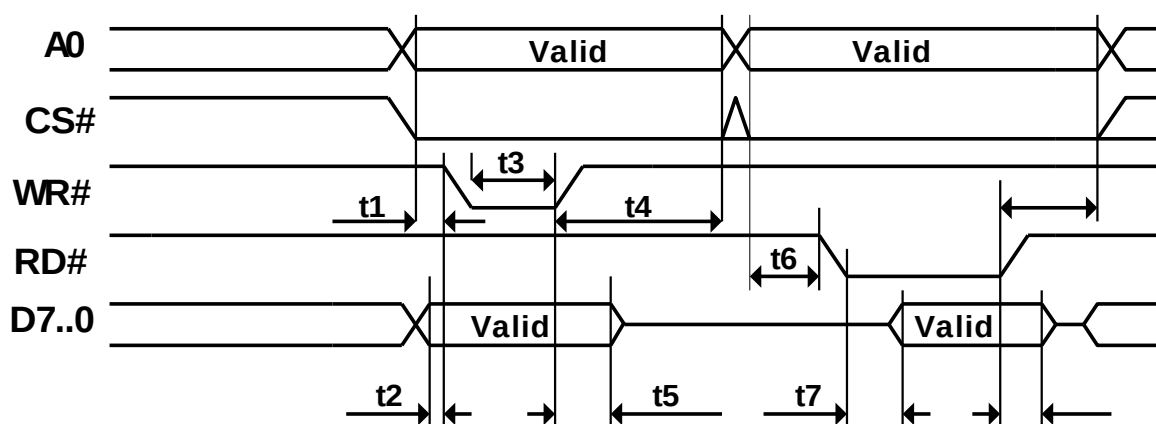


Figure 4.17 CPU-Style FIFO Interface Operation Signal Waveforms.

Name	Minimum	Typical	Maximum	Units	Description
t1	15			ns	A0/CS Setup to WR#
t2	15			ns	Data setup to WR#
t3	20			ns	WR# Pulse width
t4	5			ns	A0/CS Hold from WR#
t5	5			ns	Data hold from WR#
t6	15			ns	A0/CS Setup to RD#
t7	15		50	ns	Data delay from RD#
t8	5			ns	A0/CS hold from RD#
t9	0		30	ns	Data hold time from RD#

Table 4.9 CPU-Style FIFO Interface Operation Signal Timing.

An example of the CPU-style FIFO interface connection is shown in Figure 4.18.

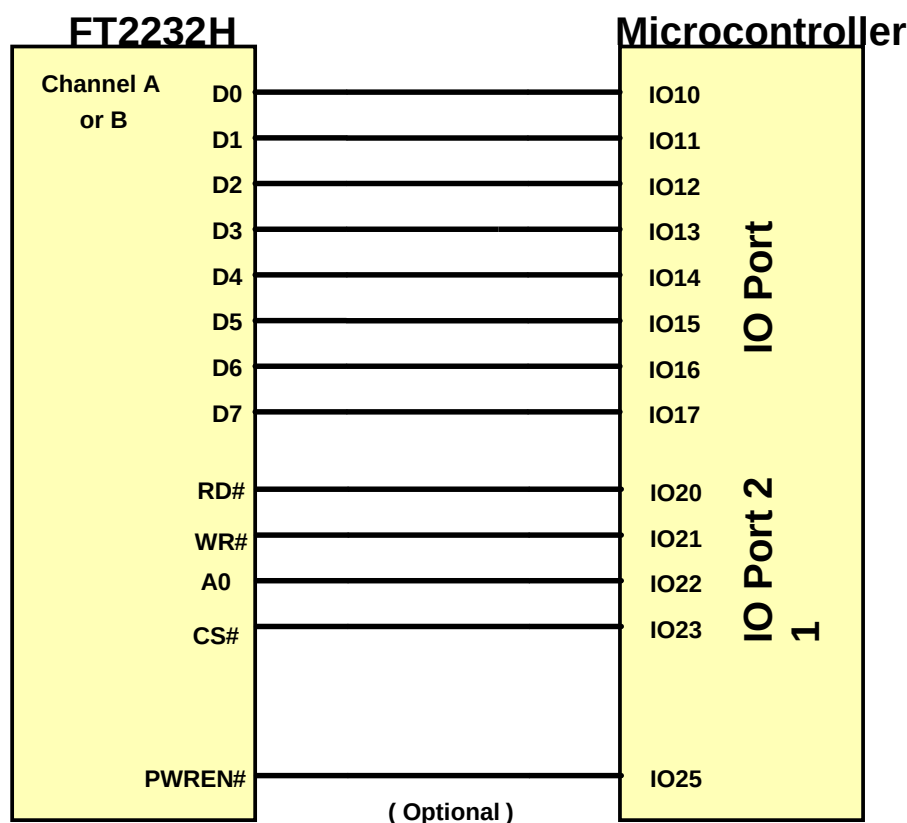


Figure 4.18 CPU-Style FIFO Interface Example

4.10 Synchronous and Asynchronous Bit-Bang Interface Mode Description

The FT2232H channel A or channel B can be configured as a bit-bang interface. There are two types of bit-bang modes: synchronous and asynchronous.

Asynchronous Bit-Bang Mode

Asynchronous Bit-Bang mode is the same as BM-style Bit-Bang mode, except that the internal RD# and WR# strobes (RDSTB# and WRSTB#) are now brought out of the device to allow external logic to be clocked by accesses to the bit-bang IO bus.

On either or both channels any data written to the device in the normal manner will be self clocked onto the data pins (those which have been configured as outputs). Each pin can be independently set as an input or an output. The rate that the data is clocked out at is controlled by the baud rate generator.

For the data to change there has to be new data written, and the baud rate clock has to tick. If no new data is written to the channel, the pins will hold the last value written.

Synchronous Bit-Bang Mode

The synchronous Bit-Bang mode will only update the output parallel port pins whenever data is sent from the USB interface to the parallel interface. When this is done, the WRSTB# will activate to indicate that the data has been read from the USB Rx FIFO buffer and written out on the pins. Data can only be received from the parallel pins (to the USB Tx FIFO interface) when the parallel interface has been written to.

With Synchronous Bit-Bang mode data will only be sent out by the FT2232H if there is space in the FT2232H USB TXFIFO for data to be read from the parallel interface pins. This Synchronous Bit-Bang mode will read the data bus parallel I/O pins first, before it transmits data from the USB Rx FIFO. It is

therefore 1 byte behind the output, and so to read the inputs for the byte that you have just sent, another byte must be sent.

For example :-

(1) Pins start at 0xFF
Send 0x55,0xAA
Pins go to 0x55 and then to 0xAA
Data read = 0xFF,0x55

(2) Pins start at 0xFF
Send 0x55,0xAA,0xAA
(repeat the last byte sent)
Pins go to 0x55 and then to 0xAA
Data read = 0xFF,0x55,0xAA

Synchronous Bit-Bang Mode differs from Asynchronous Bit-Bang mode in that the device parallel output is only read when the parallel output is written to by the USB interface. This makes it easier for the controlling program to measure the response to a USB output stimulus as the data returned to the USB interface is synchronous to the output data.

Asynchronous Bit-Bang mode is enabled using Set Bit Bang Mode driver command. A hex value of 1 will enable Asynchronous Bit-Bang mode.

Synchronous Bit-Bang mode is enabled using Set Bit Bang Mode driver command. A hex value of 4 will enable Synchronous Bit-Bang mode.

See application note [AN2232-02 Bit Mode Functions for the FT2232](#) for more details and examples of using the bit-bang modes.

An example of the synchronous bi-bang mode timing is shown in Figure 4.19.

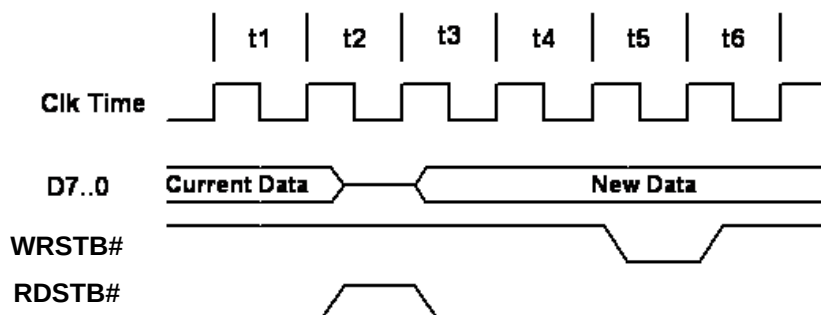


Figure 4.19 Synchronous Bit-Bang Mode Timing Interface Example

Name	Description
t1	Current pin state is read.
t2	RDSTB# is set inactive and data on the parallel I/O pins is read and sent to the USB host.
t3	RDSTB# is set active again, and any pins that are output will change to their new data.
t4	1 clock cycle to allow for data setup.
t5	WRSTB# goes active. This indicates that the host PC has written new data to the I/O parallel data pins.
t6	WRSTB# goes inactive.

Table 4.10 Synchronous Bit-Bang Mode Timing Interface Example Timings

WRSTB# = this output indicates when new data has been written to the I/O pins from the Host PC (via the USB interface).

RDSTB# = this output rising edge indicates when data has been read from the I/O pins and sent to the Host PC (via the USB interface).

The WRSTB# goes active in t4. The WRSTB# goes active when data is read from the USB RXFIFO (i.e. sent from the PC). The RDSTB# goes inactive when data is sampled from the pins and written to the USB TXFIFO (i.e. sent to the PC). The SETUP command to the FT2232H is used to setup the bit-mode. This command also contains a byte wide data mask to set the direction of each bit. The direction on each pin doesn't change unless a new SETUP command is used to modify the direction.

The WRSTB# and RDSTB# strobes is only a guide to what may be happening depending on the direction of the bus. For example if all pins are configured as inputs, it is still necessary to write to these pins in order to get the FT2232H to read those pins even though the data written will never appear on the pins.

Signals and data-flow are illustrated in Figure 4.20.

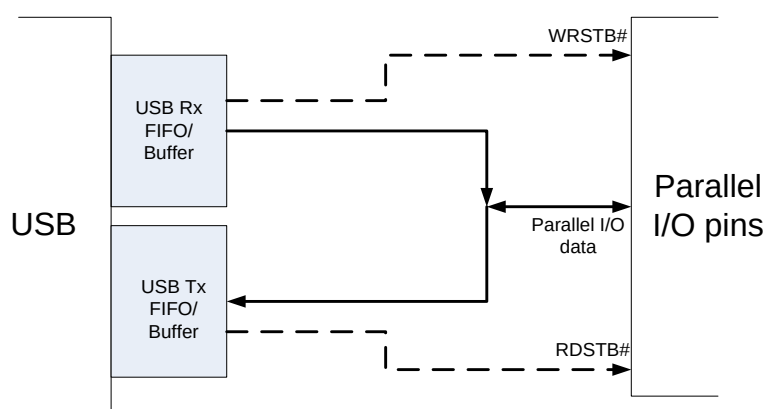


Figure 4.20 Bit-bang Mode Dataflow Illustration Diagram

4.11 RS232 UART Mode LED Interface Description

When configured in UART mode the FT2232H has two IO pins on each channel dedicated to controlling LED status indicators, one for transmitted data the other for received data. When data is being transmitted / received the respective pins drive from tri-state to low in order to provide indication on the LED's of data transfer. A digital one-shot timer is used so that even a small percentage of data transfer is visible to the end user.

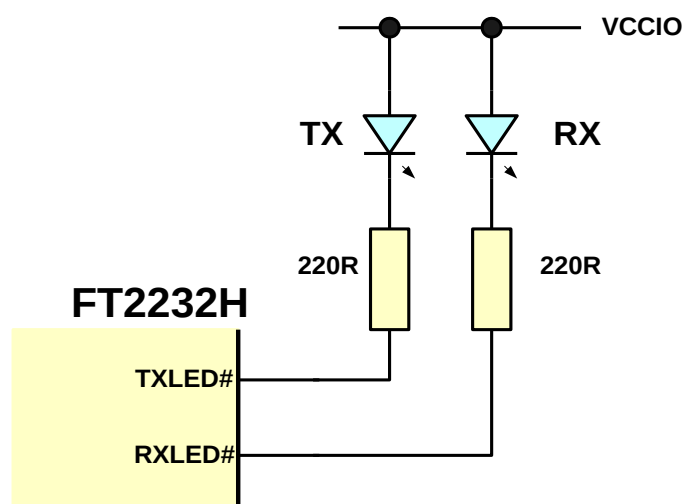


Figure 4.21 Dual LED UART Configuration

Figure 4.21 shows a configuration using two individual LED's – one for transmitted data the other for received data.

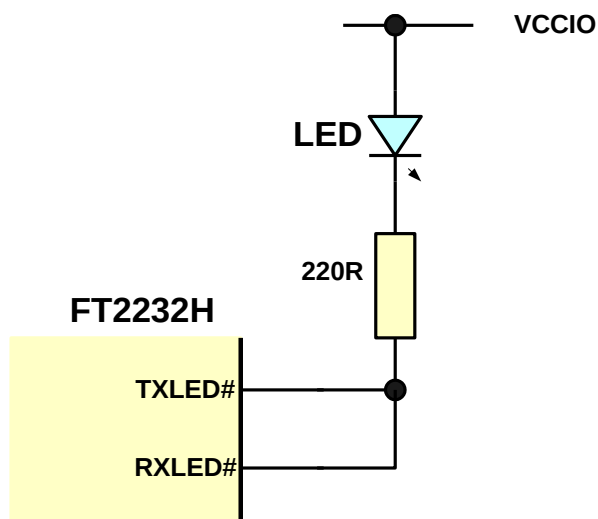


Figure 4.22 Single LED UART Configuration

In Figure 4.22, the transmit and receive LED indicators are wire-OR'ed together to give a single LED indicator which indicates any transmit or receive data activity.

Note that the LED's are connected to the same supply as VCCIO.

4.12 Send Immediate / Wake Up (SIWU#)

The SIWU# function is available in the FIFO modes and in bit bang mode.

The Send Immediate portion is used to flush data from the chip back to the PC. This can be used to get short packets of data back to the PC without waiting for the latency timer to expire.

This mechanism should only be used when you have stopped sending data to the chip to avoid overrun.

The data transfer is flagged to the USB host by the falling edge of the signal.

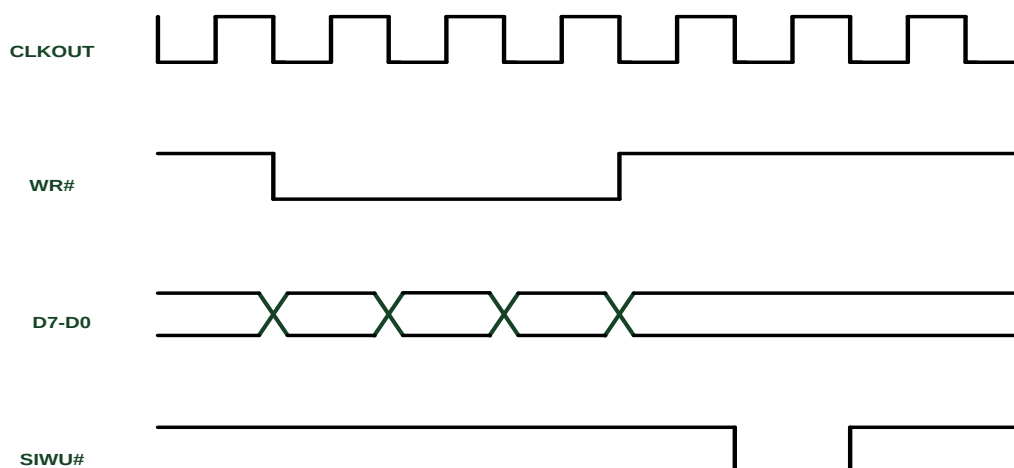


Figure 4.23 Using SIWU#

When the pin is being used for a Wake Up function to wake up a sleeping PC a 20 ms negative pulse on this pin is required. When the pin is being used to flush the buffer (Send Immediate), a 250 ns negative pulse on this pin is required.

Notes:

1. When using remote wake-up, ensure the resistors are pulled-up in suspend. Also ensure peripheral designs do not allow any current sink paths that may partially power the peripheral.
2. If remote wake-up is enabled, a peripheral is allowed to draw up to 2.5 mA in suspend. If remote wake-up is disabled, the peripheral must draw no more than 500 μ A in suspend.
3. If a Pull-down is enabled, the FT2232H will not wake up from suspend.

4.13 FT2232H Mode Selection

The 2 channels of the FT2232H reset to 2 asynchronous serial interfaces.

Following a reset the required mode of each channel is determined by the contents of the EEPROM (programmed using [FT_PROG](#)).

The EEPROM contents determine if the 2 channels have been configured as FT232 asynchronous serial interface, FT245 FIFO interface, CPU-style FIFO interface or Fast Serial Interface.

Following a reset, the EEPROM is read to determine which mode is configured. After device enumeration, an **FT_SetBitMode** command (refer to [D2XX Programmers Guide](#)) can be sent to the USB driver to switch the selected interface into the required mode – asynchronous bit-bang, synchronous bit-bang or MPSSE.

When in FT245 FIFO mode, the **FT_SetBitMode** command can be used to select either Synchronous FIFO (**FT_SetBitMode** = 0x40) or Asynchronous FIFO mode. (Note that Asynchronous FIFO mode must be selected on both channels before selecting the Synchronous FIFO mode. This means that an EEPROM is needed to initially configure Asynchronous FIFO mode before software configures the Synchronous FIFO mode).

When Synchronous FIFO mode selected, channel A uses all the memory resources of channel B. As such channel B is then not available. In this case the state of the channel B pins is determined when the configuration is switched to Asynchronous FIFO mode. If channel B had not been used for any data transfer before configuration of Asynchronous FIFO mode, then the channel B pins will remain in their default mode (D7:0=tri-stated but pulled high through 75 K resistor, TXE# =low, RXF# =high. RD# and WR# are inputs and should be pulled high). An MPSSE command, **set_data_bits** can be used to configure the channel B pins as inputs before configuring channel A as Synchronous FIFO. This avoids the channel B pins driving against any interfaces (such as SPI) which may have been configured previous to any switching of channel A to Synchronous FIFO mode. Refer to Section 3.6 Set / Read Data Bits High / Low Bytes of [AN_108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#) for the **set_data_bits** command and further information on the MPSSE used in MCU Host BUS Emulation mode.

The MPSSE can be configured directly using the D2XX commands. Refer to the [D2XX Programmers Guide](#). The application note gives further explanation and examples for the MPSSE.

4.13.1 Do I need an EEPROM?

The following Table 4.11 summarises what modes are configurable using the EEPROM or the application software.

	ASYNC Serial UART	ASYNC 245 FIFO	SYNC 245 FIFO	ASYNC Bit- bang	SYNC Bit- bang	MPSSE	Fast Serial interface	CPU- Style FIFO	Host Bus Emulation
<i>EEPROM configured</i>	YES	YES	YES				YES	YES	
<i>Application Software configured</i>			YES	YES	YES	YES			YES

Table 4.11 Configuration Using EEPROM and Application Software

5 Devices Characteristics and Ratings

5.1 Absolute Maximum Ratings

The absolute maximum ratings for the FT2232H devices are as follows. These are in accordance with the Absolute Maximum Rating System (IEC 60134). Exceeding these values may cause permanent damage to the device.

Parameter	Value	Unit
Storage Temperature	-65°C to 150°C	Degrees C
Floor Life (Out of Bag) At Factory Ambient (30°C / 60% Relative Humidity)	168 Hours (IPC/JEDEC J-STD-033A MSL Level 3 Compliant)*	Hours
Ambient Operating Temperature (Power Applied)	-40°C to 85°C	Degrees C
MTTF FT2232HL / FT2232HQ / FT2232H-56Q	9,110,463	hours
VCORE Supply Voltage	-0.3 to +2.0	V
VCCIO IO Voltage	-0.3 to +4.0	V
DC Input Voltage – USBDP and USBDM	-0.5 to +3.63	V
DC Input Voltage – High Impedance Bi-directional (CBUS & DBUS powered from VCCIO)	-0.3 to +5.8	V
DC Input Voltage – All Other Inputs	-0.5 to + (VCCIO +0.5)	V
DC Output Current – Outputs	16	mA

Table 5.1 Absolute Maximum Ratings

* If devices are stored out of the packaging beyond this time limit the devices should be baked before use. The devices should be ramped up to a temperature of +125°C and baked for up to 17 hours.

5.2 DC Characteristics

The I/O pins are +3.3 V cells, which are +5 V tolerant (except the USB PHY pins).

DC Characteristics (Ambient Temperature = -40°C to +85°C).

Parameter	Description	Minimum	Typical	Maximum	Units	Conditions
VCORE	VCC Core Operating Supply Voltage	1.62	1.80	1.98	V	
VCCIO*	VCCIO Operating Supply Voltage	2.97	3.30	3.63	V	Cells are 5 V tolerant
VREGIN	VREGIN Voltage regulator Input	3.00	3.30	3.60	V	
VREGOUT	Voltage regulator Output	1.71	1.80	1.89	V	
Ireg	Regulator Current			150	mA	VREGIN +3.3 V
Icc1	Core Operating Supply Current	---	70	---	mA	VCORE = +1.8 V Normal Operation
Icc1r	Core Reset Supply Current	---	5	---	mA	VCORE = +1.8 V Device in reset state
Icc1s	Core Suspend Supply Current		500		μA	VCORE = +1.8 V USB Suspend

Table 5.2 Operating Voltage and Current (except PHY)

NOTE: Failure to connect all VCCIO pins will result in failure of the device.

The I/O pins are +3.3 V cells, which are +5 V tolerant (except the USB PHY pins).

Parameter	Description	Minimum	Typical	Maximum	Units	Conditions
Voh	Output Voltage High	2.40	3.14		V	Ioh = +/-2 mA I/O Drive strength* = 4 mA
			3.20		V	I/O Drive strength* = 8 mA
			3.22		V	I/O Drive strength* = 12 mA
			3.22		V	I/O Drive strength* = 16 mA
Vol	Output Voltage Low		0.18	0.40	V	Iol = +/-2 mA I/O Drive strength* = 4 mA
			0.12		V	I/O Drive strength* = 8 mA
			0.08		V	I/O Drive strength* = 12 mA
			0.07		V	I/O Drive strength* = 16 mA
Vil	Input low Switching Threshold		-	0.80	V	LVTTL
Vih	Input High Switching Threshold	2.00	-		V	LVTTL
Vt	Switching Threshold		1.50		V	LVTTL
Vt-	Schmitt trigger negative going threshold voltage	0.80	1.10	-	V	
Vt+	Schmitt trigger positive going threshold voltage		1.60	2.00	V	
Rpu	Input pull-up resistance	40	75	190	KΩ	Vin = 0
Rpd	Input pull-down resistance	40	75	190	KΩ	Vin = VCCIO
Iin	Input Leakage Current	15	45	85	μA	Vin = 0
Ioz	Tri-state output leakage current		+/-10		μA	Vin = 5.5 V or 0

Table 5.3 I/O Pin Characteristics VCCIO = +3.3 V (except USB PHY pins)

* The I/O drive strength and slow slew-rate are configurable in the EEPROM.

DC Characteristics (Ambient Temperature = -40°C to +85°C).

Parameter	Description	Minimum	Typical	Maximum	Units	Conditions
VPHY, VPLL	PHY Operating Supply Voltage	3.0	3.3	3.6	V	3.3V I/O
Iccphy	PHY Operating Supply Current	---	30	60	mA	High-speed operation at 480 MHz
Iccphy (susp)	PHY Operating Supply Current	---	10	50	μA	USB Suspend

Table 5.4 PHY Operating Voltage and Current

Parameter	Description	Minimum	Typical	Maximum	Units	Conditions
Voh	Output Voltage High	VCORE-0.2			V	
Vol	Output Voltage Low			0.2	V	
Vil	Input low Switching Threshold		-	0.8	V	
Vih	Input High Switching Threshold	2.0	-		V	

Table 5.5 PHY I/O Pin Characteristics

5.3 ESD Tolerance

ESD protection for FT2232H IO's.

Parameter	Reference	Minimum	Typical	Maximum	Units
Human Body Model (HBM)	JEDEC EIA/JESD22-A114-B, Class 2		±2 kV		kV
Machine Mode (MM)	JEDEC EIA/JESD22-A115-A, Class B		±200 V		V
Charge Device Model (CDM)	JEDEC EIA/ JESD22-C101-D, Class-III		±500 V		V
Latch-up	JESD78, Trigger Class-II		±200 mA		mA

Table 5.6 ESD Tolerance

5.4 Thermal Characteristics

Parameter	Minimum	Typical	Maximum	Units
θ _{JA} (FT2232HL)		37.66		°C/W
θ _{JC} (FT2232HL)		8.39		°C/W
θ _{JA} (FT2232HQ)		29.67		°C/W
θ _{JC} (FT2232HQ)		14.12		°C/W
T _J (FT2232HL/FT2232HQ)	-40	25	125	°C

Table 5.7 Thermal Characteristics

Bus Powered Application example 2: Bus powered configuration (with additional 1.8 V LDO voltage regulator for VCORE).

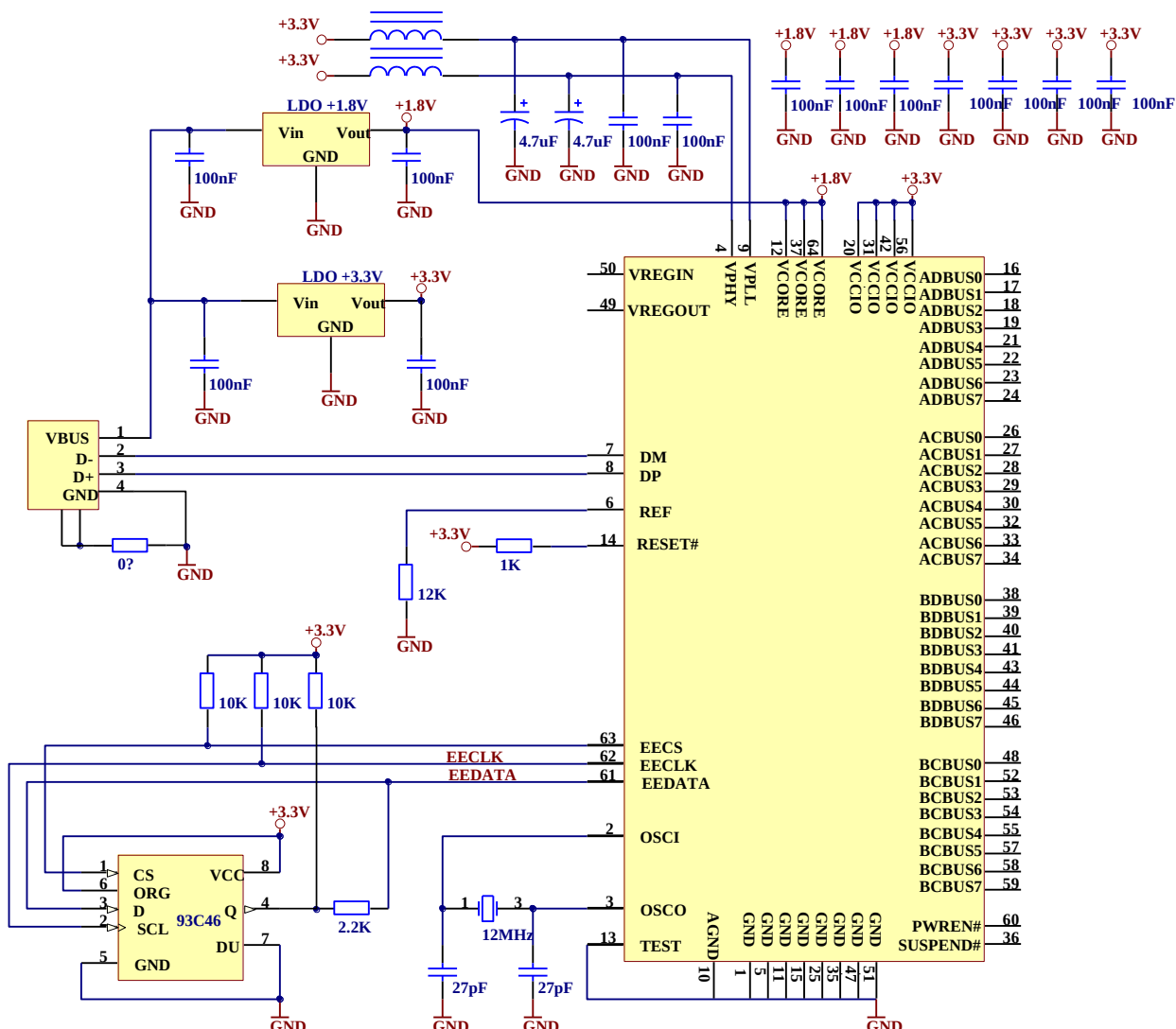


Figure 6.2 Bus Powered Configuration Example 2

Figure 6.2 illustrates the FT2232H in a typical USB bus powered configuration similar to Figure 6.1. The difference here is that the +1.8 V for the FT2232H core (VCORE) has been regulated from the VBUS as well as the +3.3 V supply to the VPLL, VPHY, VCCIO and VREGIN.

Self-Powered application example 2: Self powered configuration (with additional 1.8 V LDO voltage regulator for VCORE).

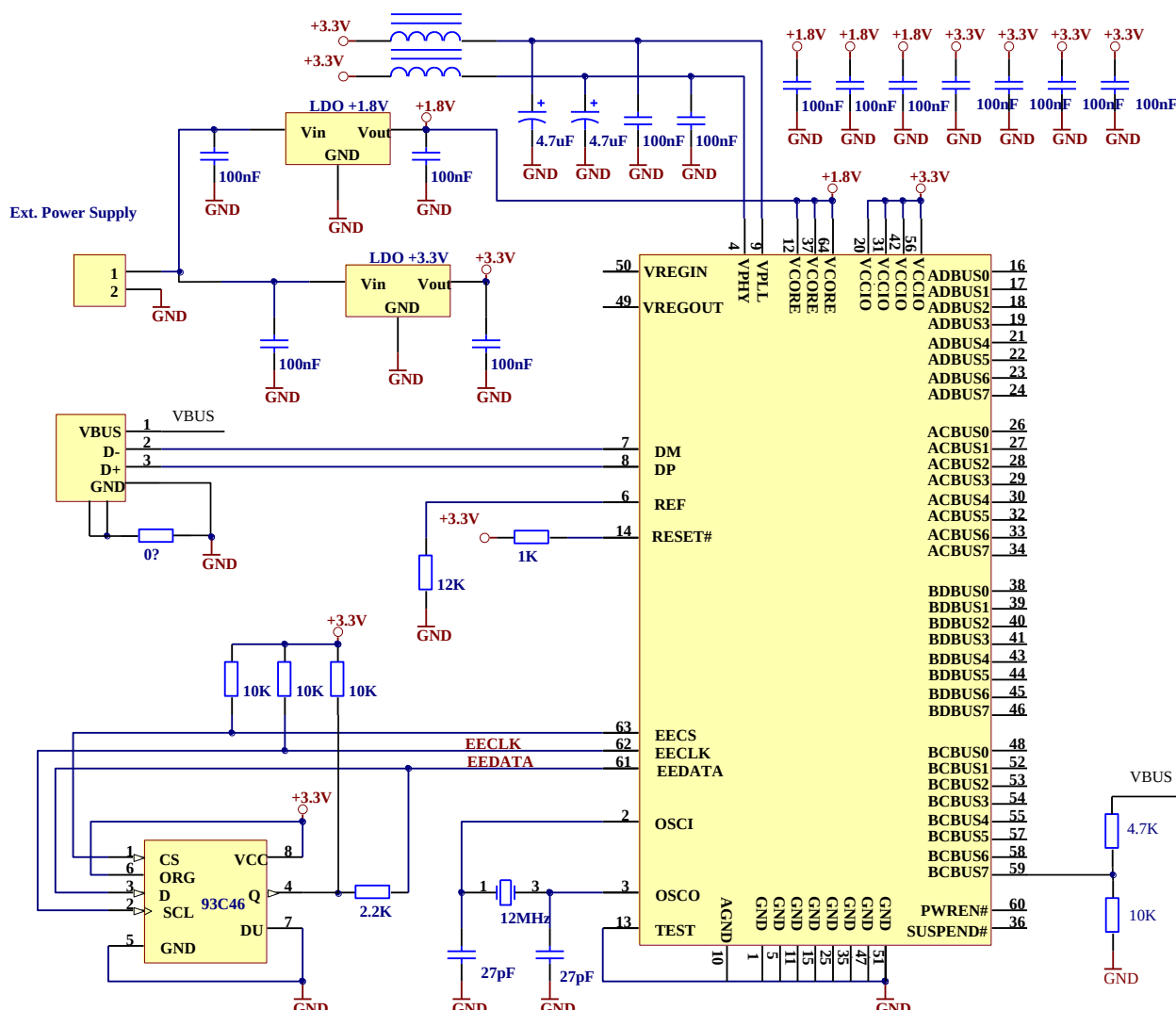


Figure 6.4 Self Powered Configuration Example 2

Figure 6.4 illustrates the FT2232H in a typical USB self-powered configuration similar to Figure 6.3. The difference here is that the +1.8 V for the FT2232H core has been regulated from the external power supply.

Note that in this set-up, the EEPROM should be configured for self-powered operation and the option "suspend on DBUS7 low" selected in [FT_PROG](#). Also this configuration uses the pin BCBUS7, so this assumes that MPSSE mode is not selected.

6.3 Oscillator Configuration

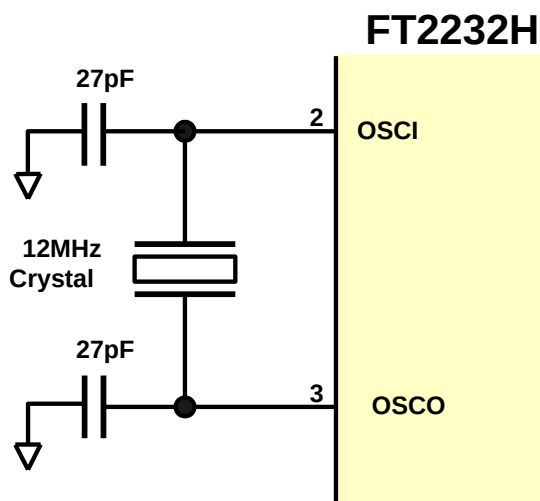


Figure 6.5 Recommended FT2232H Crystal Oscillator Configuration

Figure 6.5 illustrates how to connect the FT2232H with a 12 MHz $\pm 0.003\%$ crystal. In this case loading capacitors should be added between OSCI, OSCO and GND as shown. A value of 27 pF is shown as the capacitor in the example – this will be good for many crystals but it is recommended to select the loading capacitor value based on the manufacturer's recommendations wherever possible. It is recommended to use a parallel cut type crystal.

It is also possible to use a 12 MHz Oscillator with the FT2232H. In this case the output of the oscillator would drive OSCI, and OSCO should be left unconnected. The oscillator must have a CMOS output drive capability.

Parameter	Description	Minimum	Typical	Maximum	Units	Conditions
OSCI Vin	Input Voltage	2.97	3.30	3.63	V	
FIn	Input Frequency		12		MHz	+/- 30ppm
Ji	Cycle to cycle jitter		< 150		pS	

Table 6.1 OSCI Input characteristics

7 EEPROM Configuration

If an external EEPROM is fitted (93LC46/56/66) it can be programmed over USB using [FT_PROG](#). The EEPROM must be 16 bits wide and capable of working at a VCC supply of +3.0 to +3.6 volts.

Adding an external EEPROM allows the chip to be configured as FT232 asynchronous serial interface, FT245 FIFO interface, CPU-style FIFO interface or Fast Serial Interface.

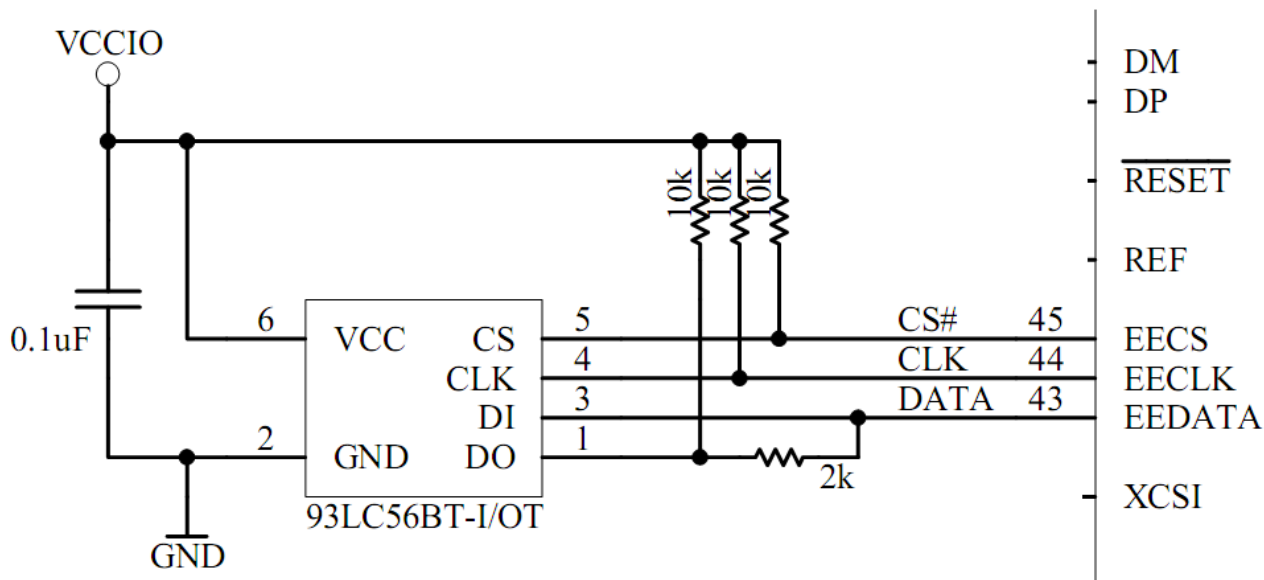


Figure 7.1 EEPROM Interface

The external EEPROM can also be used to customise the USB VID, PID, Serial Number, Product Description Strings and Power Descriptor value of the FT2232H for OEM applications. Other parameters controlled by the EEPROM include Remote Wake Up, Soft Pull Down on Power-Off and I/O pin drive strength.

If the FT2232H is used without an external EEPROM the chip defaults to a USB to FT232 asynchronous serial interface port device. If no EEPROM is connected (or the EEPROM is blank), the FT2232H uses its built-in default VID (0403), PID (6010) Product Description and Power Descriptor Value. In this case, the device will not have a serial number as part of the USB descriptor.

7.1 Default EEPROM Configuration

The external EEPROM (if it is fitted) can be programmed over USB using [FT_PROG](#). This allows a blank part to be soldered onto the PCB and programmed as part of the manufacturing and test process. Users who do not have their own USB Vendor ID but who would like to use a unique Product ID in their design can apply to FTDI for a free block of unique PIDs.

See technical note [TN_100 USB Vendor ID/Product ID Guidelines](#) for more information.

Parameter	Value	Notes
USB Vendor ID (VID)	0403h	FTDI default VID (hex)
USB Product ID (PID)	6010h	FTDI default PID (hex)
bcd Device	0700h	
Serial Number Enabled?	Yes	
Serial Number	None	
Pull down I/O Pins in USB Suspend	Disabled	Enabling this option will make the device pull down on the UART interface lines when in USB suspend mode (PWREN# is high).
Manufacturer Name	FTDI	
Product Description	Dual RS232-HS	
Max Bus Power Current	500mA	
Power Source	Bus Powered	
Device Type	FT2232H	
USB Version	0200h	Returns USB 2.0 device description to the host.
Remote Wake Up	Disabled	Taking RI# low will wake up the USB host controller from suspend in approximately 20 ms. If enabled.
Hardware Interface	UART	Allows the user to select the hardware mode of the device. Options include: UART, 245 FIFO, CPU 245, OPTO Isolate.
Suspend ACBus7 Low	Disabled	Enters low power state on ACBus7.
High Current I/Os	Disabled	Enables the high drive level on the UART and ACBUS I/O pins.
Load VCP Driver	Enabled	Makes the device load the VCP driver interface for the device.

Table 7.1 Default Configuration with a blank/no EEPROM

8 Package Parameters

The FT2232H is available in two different packages. The FT2232HL is the LQFP-64 option and the FT2232HQ is the QFN-64 package option. The solder reflow profile for both packages is described in Section 8.4. See technical note [TN_166 FTDI Example IC Footprints](#) for PCB footprint guidelines.

8.1 FT2232HQ, QFN-64 Package Dimensions

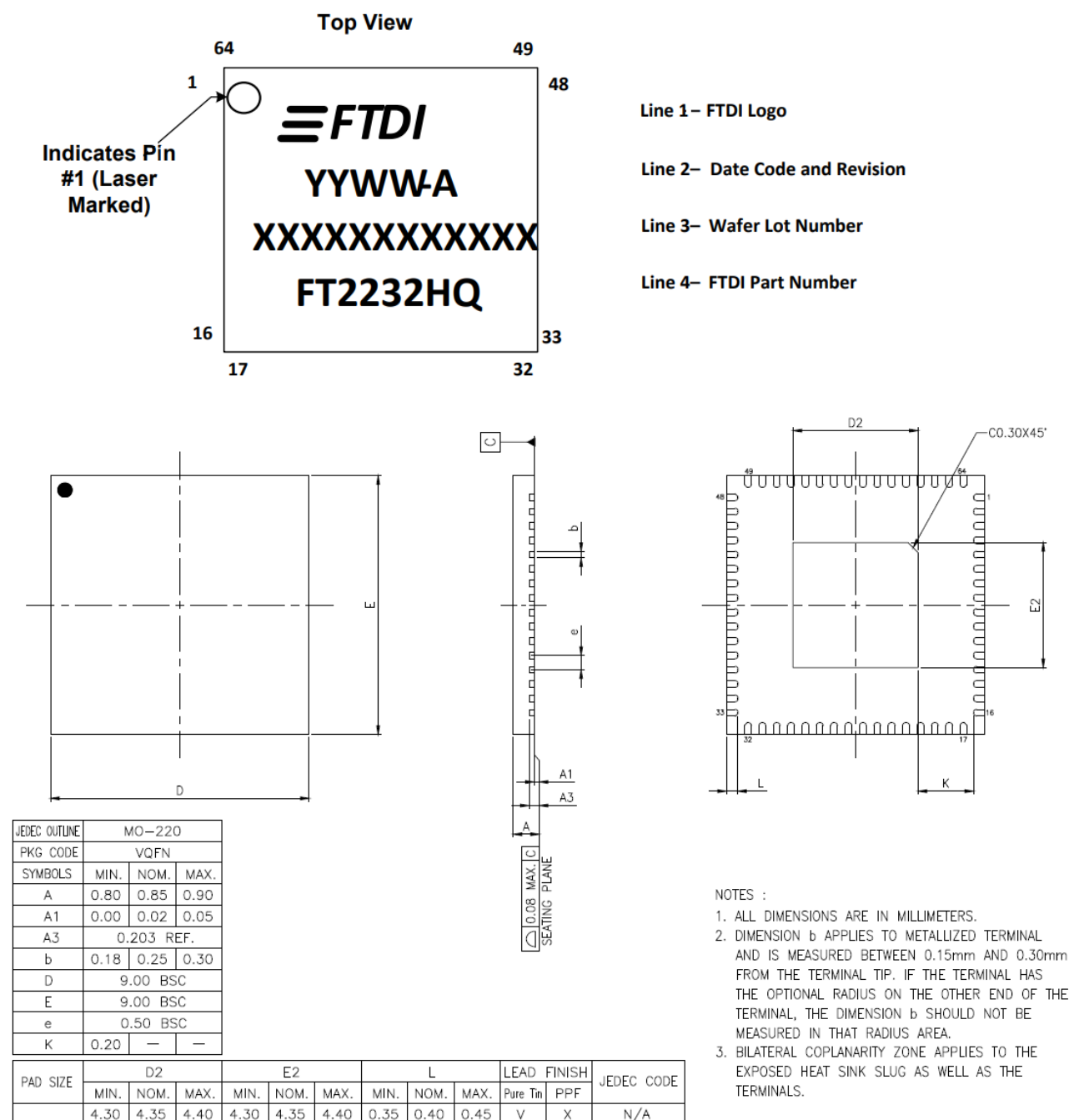


Figure 8.1 64 pin QFN Package Details

Notes:

- All dimensions are in mm.
- The bottom side central solder pad must be connected to the ground of the system.

8.2 FT2232HL, LQFP-64 Package Dimensions

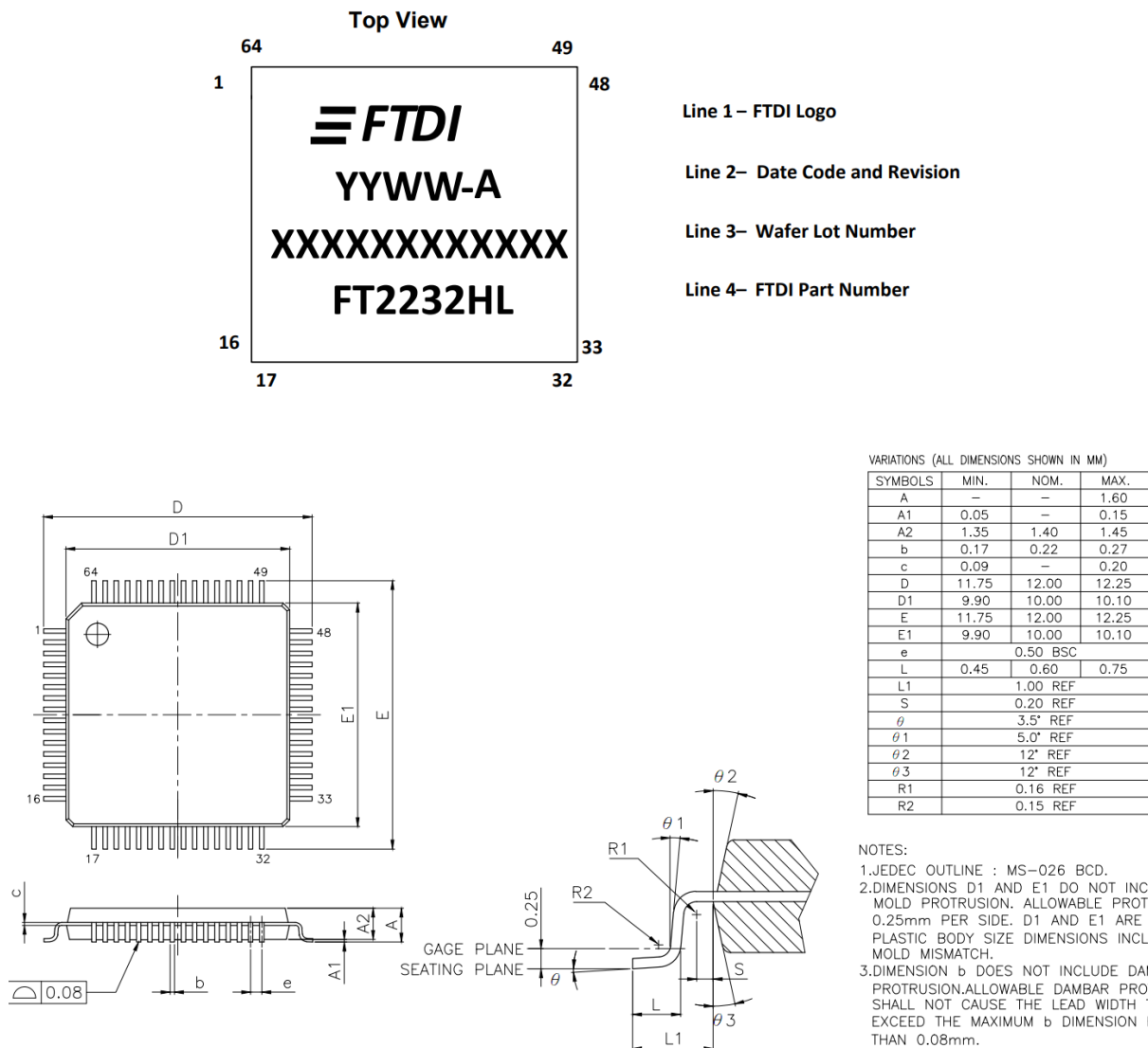


Figure 8.2 64 pin LQFP Package Details

8.3 FT2232H-56Q, VQFN-56 Package Dimensions

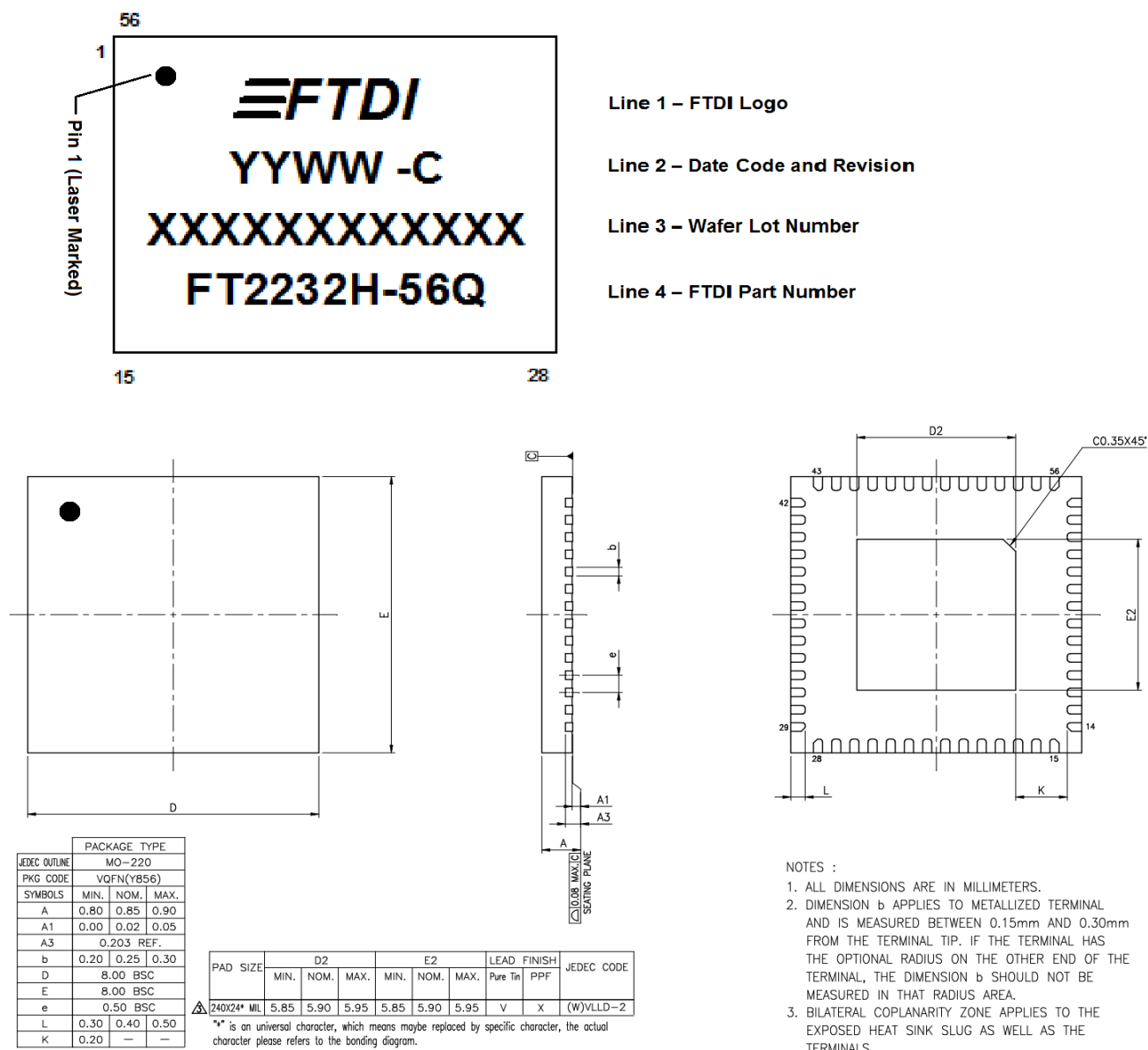


Figure 8.3 56-pin VQFN Package Details for FT2232H-56Q

Note:

The internal ground of the device is connected to the bottom side central solder pad whose dimension is 5.90 x 5.90 mm. This central solder pad must be connected to the ground of the system.

8.4 Solder Reflow Profile

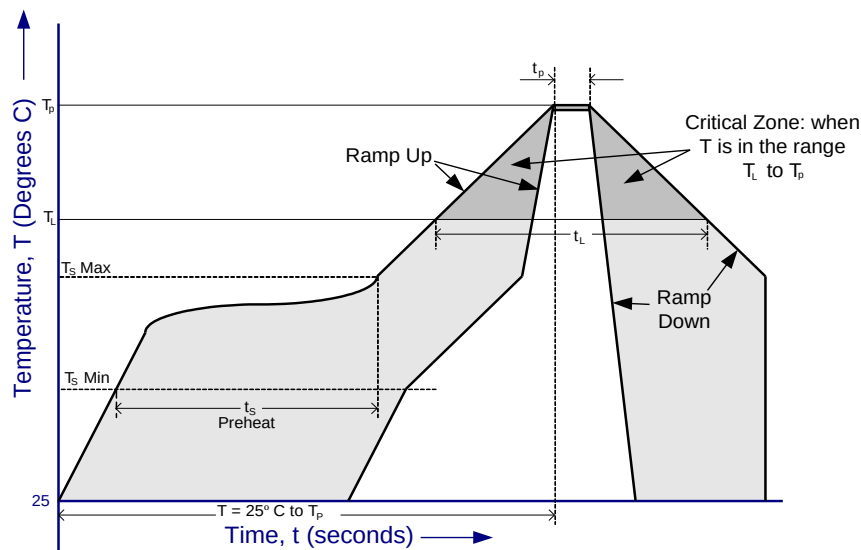


Figure 8.4 FT2232H Solder Reflow Profile

Profile Feature	Pb Free Solder Process (green material)	SnPb Eutectic and Pb free (non green material) Solder Process
Average Ramp Up Rate (T_s to T_p)	3°C / second Max.	3°C / Second Max.
Preheat - Temperature Min (T_s Min.) - Temperature Max (T_s Max.) - Time (t_s Min to t_s Max)	150°C 200°C 60 to 120 seconds	100°C 150°C 60 to 120 seconds
Time Maintained Above Critical Temperature T_L : - Temperature (T_L) - Time (t_L)	217°C 60 to 150 seconds	183°C 60 to 150 seconds
Peak Temperature (T_p)	260°C	see Table 8.2
Time within 5°C of actual Peak Temperature (T_p)	30 to 40 seconds	20 to 40 seconds
Ramp Down Rate	6°C / second Max.	6°C / second Max.
Time for $T = 25^\circ\text{C}$ to Peak Temperature, T_p	8 minutes Max.	6 minutes Max.

Table 8.1 Reflow Profile Parameter Values

SnPb Eutectic and Pb free (non green material)		
Package Thickness	Volume mm3 < 350	Volume mm3 >=350
< 2.5 mm	235 +5/-0 deg C	220 +5/-0 deg C
≥ 2.5 mm	220 +5/-0 deg C	220 +5/-0 deg C
Pb Free (green material) = 260 +5/-0 deg C		

Table 8.2 Package Reflow Peak Temperature

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Appendix A – References

Document References

[AN2232-02 Bit Mode Functions for the FT2232](#)

[AN_108 Command Processor for MPSSE and MCU Host Bus Emulation Modes](#)

[AN_110 Programmers Guide for High Speed FTCJTAG DLL](#)

[AN_111 Programmers Guide for High-Speed FTCSPI DLL](#)

[AN_113 Interfacing FT2232H Hi-Speed Devices to I2C Bus](#)

[AN_114 Interfacing FT2232H Hi-Speed Devices To SPI Bus](#)

[AN_177 User Guide for LibMPSSE-I2C](#)

[AN_135 MPSSE Basics](#)

[TN_100 USB Vendor ID/Product ID Guidelines](#)

[TN_104 Guide to Debugging Customers Failed Driver Installation](#)

[TN_166 FTDI Example IC Footprints](#)

[FT_PROG EEPROM Programming Utility](#)

Acronyms and Abbreviations

Terms	Description
CDM	Charge Device Model
CMOS	Complementary Metal Oxide Semiconductor
ESD	Electrostatic Discharge
EHCI	Extensible Host Controller Interface
EEPROM	Electrically Erasable Programmable Read-Only Memory
FIFO	First In First Out
FPGA	Field-Programmable Gate Array
HBM	Human Body Model
IC	Integrated Circuit
I ² C	Inter Integrated Circuit
JTAG	Joint Test Action Group
LDO	Low Drop Out
LED	Light Emitting Diode
LQFP	Low profile Quad Flat Package
MM	Machine Mode

Terms	Description
MCU	Microcontroller Unit
MPSSE	Multi-Protocol Synchronous Serial Engine
OHCI	Open Host Controller Interface
PLD	Programmable Logic Device
QFN	Quad Flat No-Lead
SPI	Serial Peripheral Interface
USB	Universal Serial Bus
UART	Universal Asynchronous Receiver/Transmitter
UHCI	Universal Host Controller Interface
UTMI	Universal Transceiver Macrocell Interface
VCP	Virtual COM Ports
VQFN	Very Thin Quad Flat Non-Leaded Package

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Appendix C – Revision History

Document Title: FT2232H Dual High Speed USB to Multipurpose UART/FIFO IC
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Revision	Changes	Date
1.0	Initial Release.	2008-11-04
1.10	QFN Package updated.	2008-11-20
2.01	Corrections made to table 3.6, 3.7, table on page 8. Changed description of WRSTRB# and RDSTRB#.	2009-02-01
2.02	Added note that HBE mode only operates at 60MHz. Corrections made to tray QFN 160 changed to 260. Correction made to 3.4.2, falling changed to rising.	2009-03-01
2.03	Corrections made to TxLED and RxLED pin connections. Corrected signals in Figure 4.16. Corrected signal names in Fig 2.1. Added reference to AN_108, AN_109, AN_110, AN_111 and AN_113.	2009-05-19
2.04	Added paragraph on latency timer to section 4.1.	2009-06-03
2.05	Corrected Figures 6.2, 6.3 and 6.4 – missing regulators and better way of holding self-powered designs in reset if not connected to USB. Corrected Max DC inputs on "DC Input Voltage. "All Other Inputs" pins from VCORE+0.5V to VCCIO+0.5V.	2009-06-17
2.06	Added explanation of SIWU (4.12). Added explanation of MPSSE Adaptive clocking (4.6.1). Corrected 12MHz crystal specification. Added # to TXLED, RXLED on table 3.4. Corrected TX_LED and RX_LED connections on Fig 4.1.	2009-09-21
2.07	Edited Table 3.11, references AN2232L-1 to AN_108 Updated and formatted contact information.	2010-03-12
2.08	Added TID number (Section 1.3). Added ESD specifications.	2010-05-24

Revision	Changes	Date
2.09	Corrected 'WR' to 'WR#' throughout the datasheet. Edited table 4.1 (T8 and T13 Comments). Edited section 4.7.1 and 4.7.2. Section 4.12, added clarifications about Wake up. Clarified unsupported baud rates of 7,9,10 and 11 Mbaud.	2010-09-02
2.10	Updated section 4.5, FT245 Asynchronous FIFO. Interface timing diagram. Edited section 4.3.2, 4.3.3, figure 4.2 and 4.3.	2010-11-22
2.11	Edited section 4.7. From Bit A18 to A8. Edited table 3.4 Pin 29 and 30 Description. Added feedback links.	2012-03-07
2.20	Updated 245 FIFO Asynchronous Timing Table 4.2, Figure 4.5 and 4.6.	2012-04-09
2.21	Update performance of FT245 Sync FIFO mode. Updated Table 4.1, SIWU# timing updated.	2012-06-21
2.22	Added thermal characteristics – new section 5.4. Updated FTDI USA address. Added notes details for QFN package. Added clarification for which signals are 5V tolerant.	2013-01-04
2.3	Add information for new package 56-pin VQFN.	2016-03-18
2.4	Added section Default EEPROM Configuration.	2016-06-03
2.5	Clarified default config is without an EEPROM/balnk EEPROM.	2016-07-07
2.6	Updated the following pictures – Figure 8.1, 8.2 and 8.3; Removed table 8.1 (as is now part of the figure); Updated the notes under fig.8.1; Updated the notes under fig.8.3.	2019-05-27
2.7	Updated CPU-Style FIFO wake-up details in section 3.1.2, 3.2.2 and 4.9. In this mode, wake-up is requested by the states of ACBUS1/ACBUS1/ACBUS3 instead of by SI/WU#. In section 5.2 Table 5.2 changed minimum value from 3.3V to 3.0V.	2021-06-10
2.8	Release version to add and incorporate changes from previous version 2.7 (as per above). Update Driver Support. Update reference link and some document format.	15-05-2024

Revision	Changes	Date
2.9	Deleted repeated paragraph on page 31. The last sentence in the first paragraph on page 27 implies that VREGOUT is to be connected to VREGIN. This was changed to VCORE. Added MTTF information to Table 5.1. Changed heading name on section 3.2.1 to FT2232H-56Q. Updated field t9 in table 4.2 to read "DATA hold time after WR# goes active"	06-09-2024